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Rare Disease Lead Scientist
North East and Yorkshire
Genomic Medicine Service

It's all about

you & me & him & her & them.



Clinical genomic testing can happen at any point in lifetime and for multiple different health reasons

Prenatal testing

Reproductive choices/PGD

Paediatric disorders

Hereditary cancer testing

Newborn screening

Late-onset disorders



GUIDELINE

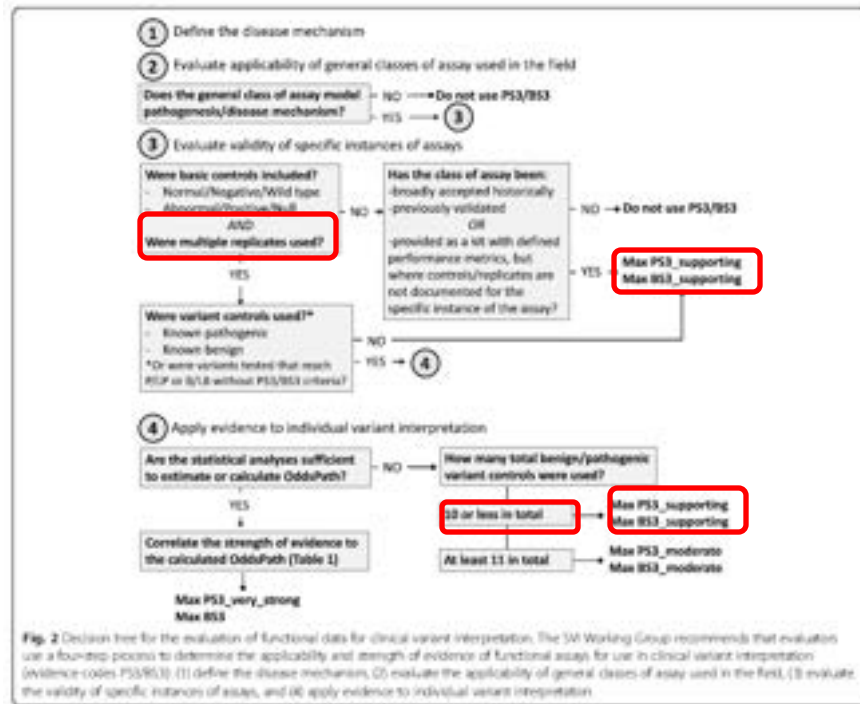
Open Access

Recommendations for application of the functional evidence PS3/BS3 criterion using the ACMG/AMP sequence variant interpretation framework



Sarah E. Brnich¹ , Ahmad N. Abou Tayoun², Fergus J. Couch³, Garry R. Cutting⁴, Marc S. Greenblatt⁵, Christopher D. Heinen⁶, Dona M. Kanavy³, Xi Luo², Shannon M. McNulty¹, Lea M. Starita^{8,9}, Sean V. Tavtigian¹⁰, Matt W. Wright¹¹, Steven M. Harrison¹², Leslie G. Biesecker¹³, Jonathan S. Berg¹⁴ and On behalf of the Clinical Genome Resource Sequence Variant Interpretation Working Group

- Most traditional functional assays only reach PS3_sup
- How to assess classification of controls used?
- High workload involved in Brnich-evaluation of each assay and many variables



Cancer Variant Interpretation Group (CanVIG) MAVE survey results

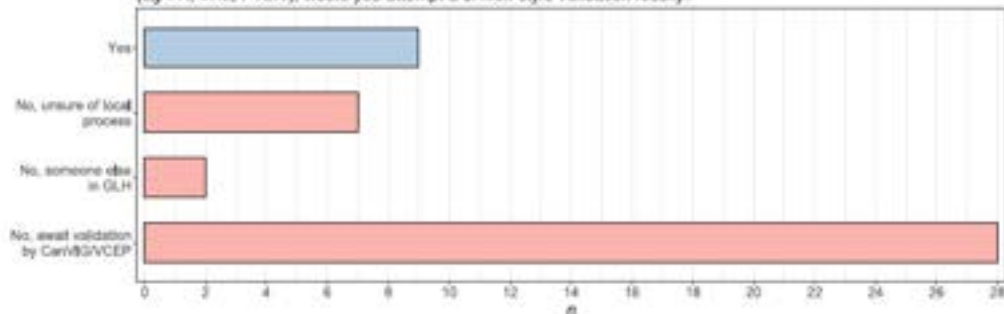
ARTICLE

Validating data from multiplex assays of variant effect: A CanVIG-UK national survey of NHS clinical scientists

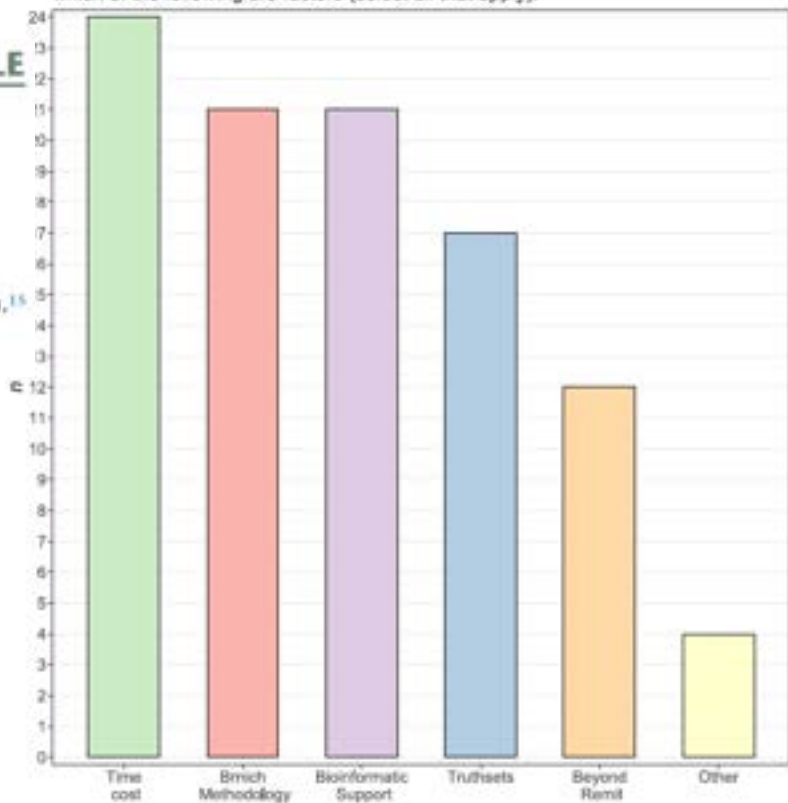
Sophie Allen,¹ Alice Garrett,^{1,2} Charlie F. Rowlands,¹ Miranda Durkie,³ George J. Burghel,^{4,5} Rachel Robinson,⁶ Alison Callaway,⁷ Joanne Field,⁸ Bethan Frugtniet,² Sheila Palmer-Smith,⁹ Jonathan Grant,¹⁰ Judith Pagan,¹¹ Trudi McDevitt,¹² Katie Snape,² Helen Hanson,^{13,14} Terri McVeigh,¹⁵ David J. Adams,¹⁶ Gregory M. Findlay,¹⁷ Reh The American Journal of Human Genetics 112, 1479–1488, June 5, 2025
Clare Turnbull,^{1,15,*} and CanVIG-UK Consortium

B

If there was a new MAVE published that was useful for a gene tested by your GLH that you report on (eg FH, VHL, PTEN), would you attempt a Brnich-style validation locally?



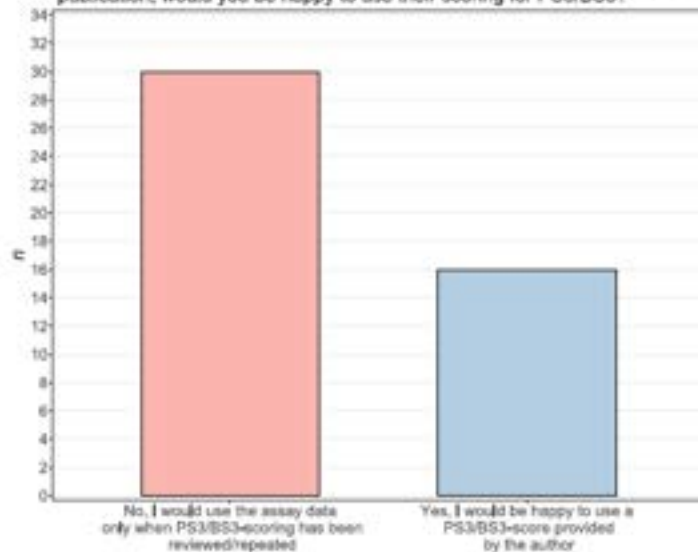
If you answered No to the previous question about conducting a Brnich-style validation, which of the following are factors (select all that apply):



CanVIG MAVE survey results

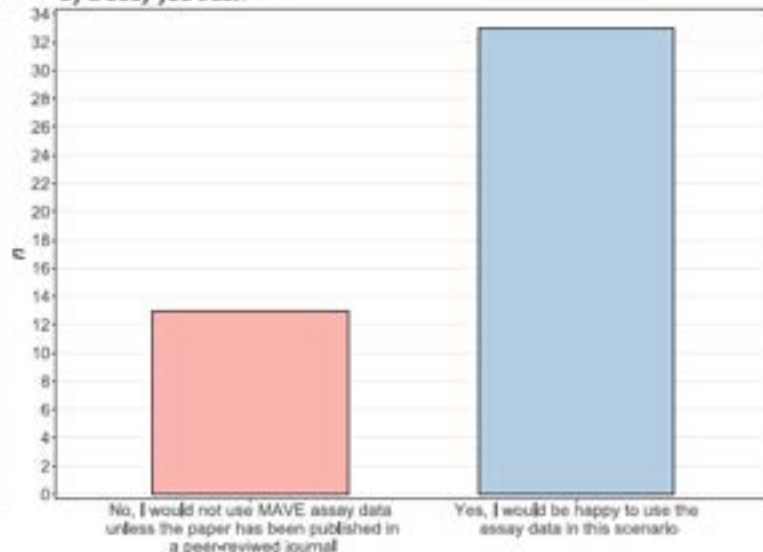
A

If a newly published MAVE assay included a Brnrich-style validation in the publication, would you be happy to use their scoring for PS3/BS3?



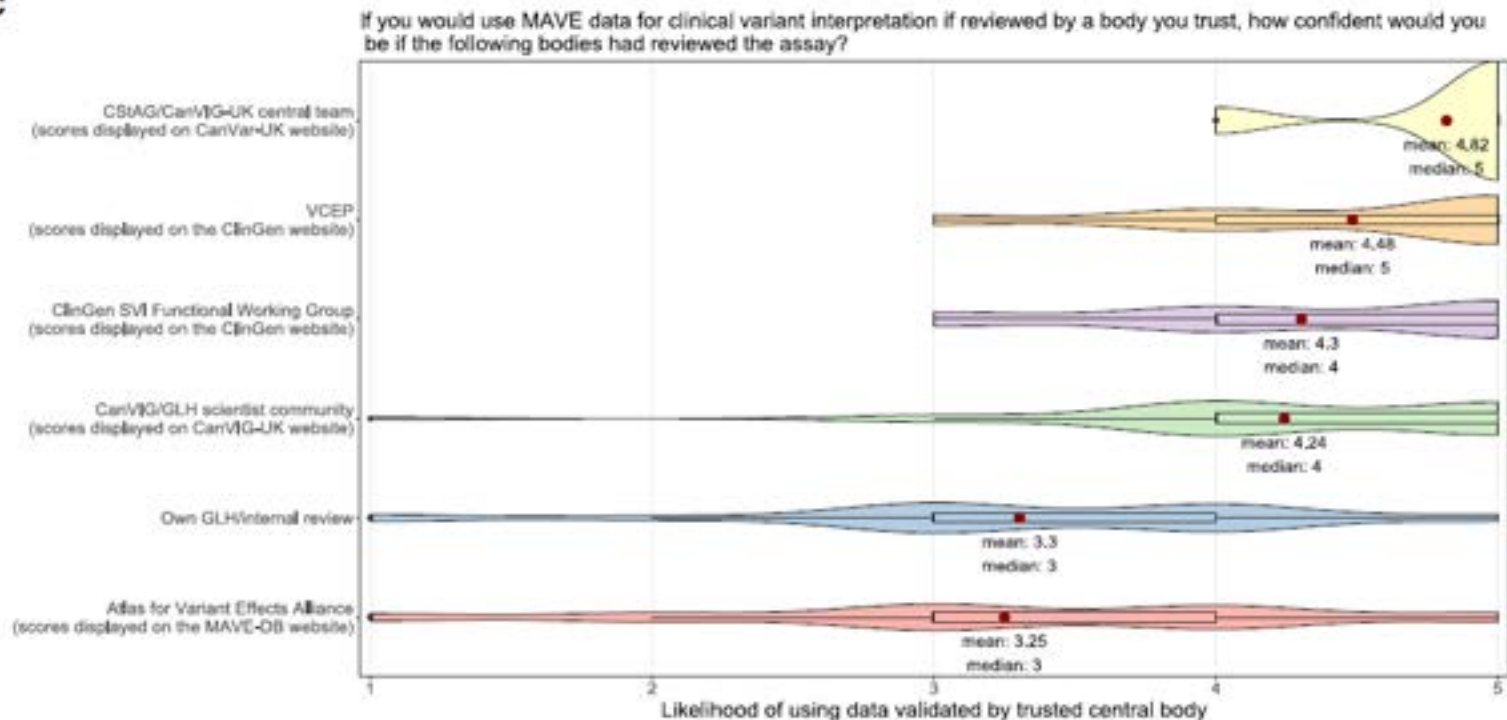
B

If a new MAVE assay was not yet published, but the paper was available on a pre-print server (eg bioRxiv), would you be happy to use the data for clinical variant interpretation if the MAVE assay had been reviewed by a body you trust?



CanVIG MAVE survey results

C



CanVIG survey results

“Clinical scientists are stretched too far already”

“We haven’t defined a formal process to undertake this activity, we would do a superficial review using the Brinch flow chart [sic] rather than a specific assessment”

“It’s not the most intuitive calculation and would be very time-consuming to do”

“no idea where to start”

“We need extra funding for personnel to perform this task”

CanGene
CanVar

Contact us

CaseyMS-100

CarMax-UK

National Cancer Genetics Act (NCGL)

Notes and Equations

CanVIG-UK Functional Studies Scoring (live sync)

[Access Full Scoring Sheet](#)

Summary of functional studies reviewed and scored by CarIS-UK in accordance to Clinical Genome Resource (CGR) recommendations (Brich et al. 2020)

[Click 'Access Full Scoring Sheet' to see the full summary in Google Sheets.](#)

[illegible]

Real Clinical Cases



Case 1: child with foramen magnum meningioma aged 8

Absent gnomAD v4.1 (PM2)

REVEL 0.957/HCI prior 0.90 (PP3)

ClinVar: VUS x2

InSiGHT and CanVIG: 3 points



The Programme

CanVIG-UK

CanVar-UI

	Gene	Reference	PMID	Evidence Weighting	
				PS3	BS3
1	MSH2	Jia et al, 2021	33357406	Strong	Strong
2					
3					
21					

ARTICLE

Massively parallel functional testing of *MSH2* missense variants conferring Lynch syndrome risk

Xiaoyan Jia,¹ Bala Bharathi Burugula,¹ Victor Chen,¹ Rosemary M. Lemons,¹ Sajini Jayakody,¹ Mariam Maksutova,¹ and Jacob O. Kitman^{1,2,*}

The American Journal of Human Genetics 108, 163–175, January 7, 2021

MSH2 c.2345C>T p.(Thr782Ile)

Summary

In Silico Predictions

Control / Case Frequency

Functional Analysis

[Kitman et al. Am J Hum Genet. 2021](#)

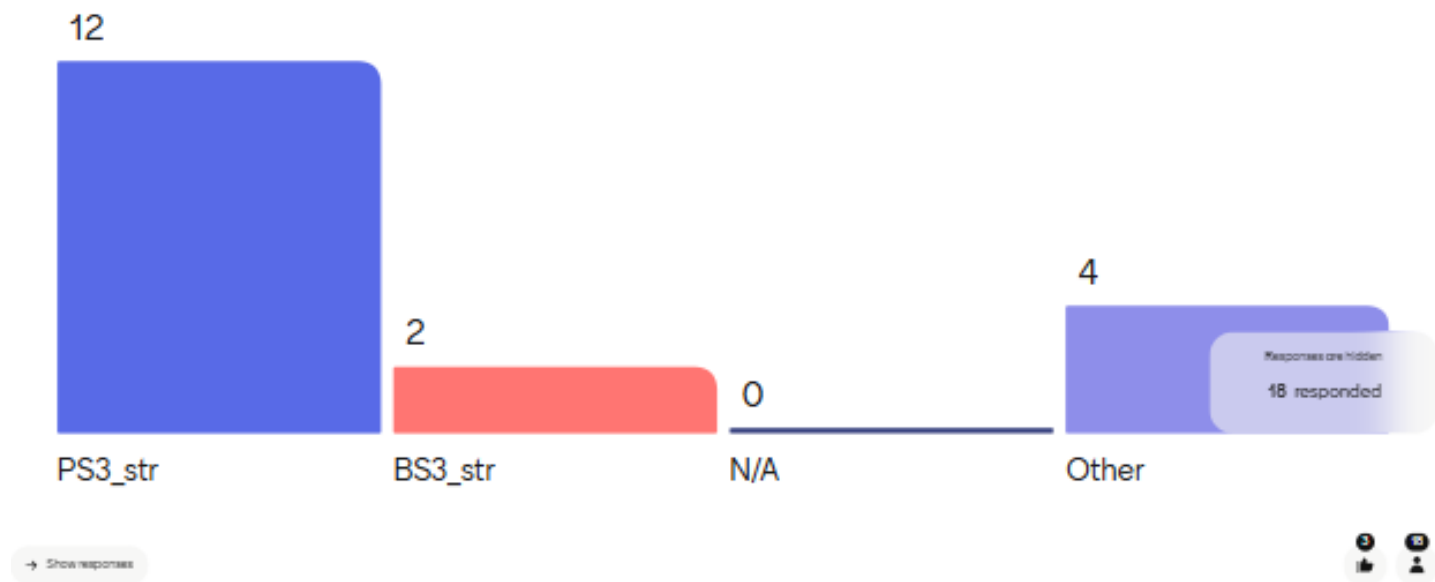
Loss-of Function Score

0.06

Loss-of Function Classification

LOF

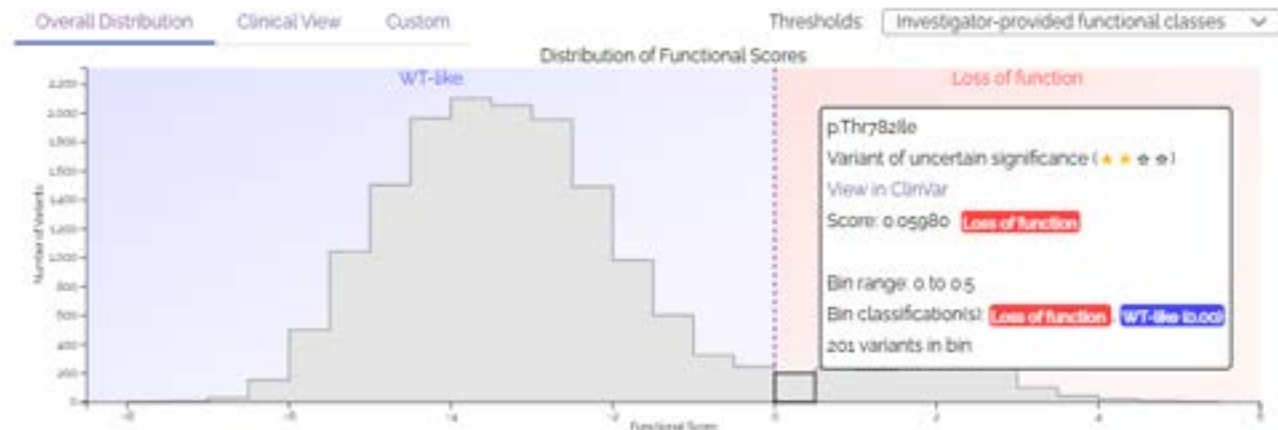
Case 1: How would you score this functional data?



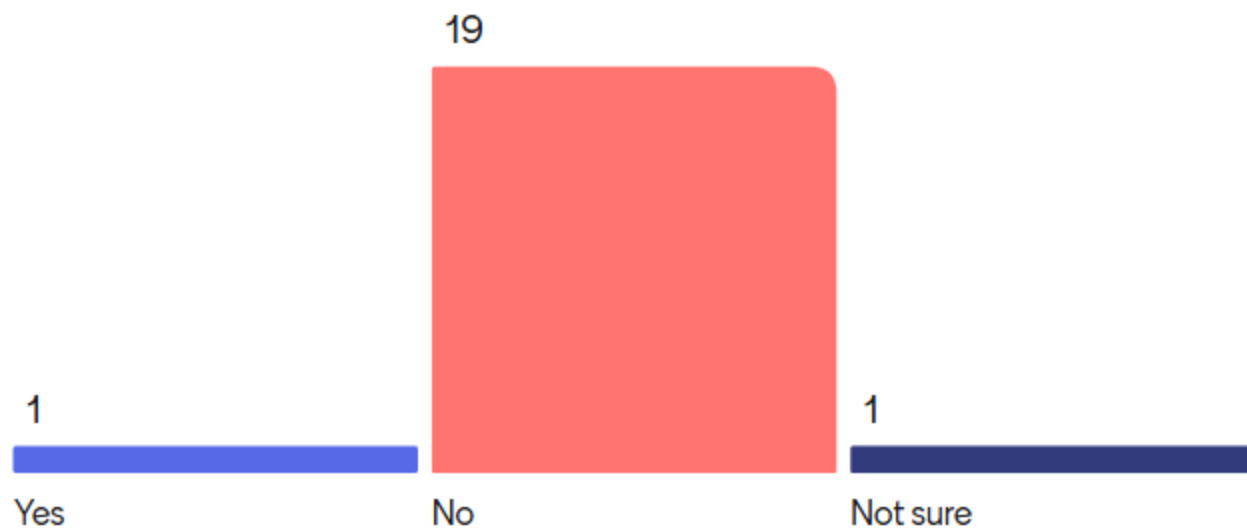
MaveMD

Variant: NP_000242.1:p.Thr782Ile		Jia X <i>et al</i> MSH2 2021 <i>Am J Hum Genet</i>	
ClinGen allele ID	P457992876	Gene (HGNC symbol)	MSH2
ClinVar allele ID	s548048	Assay Type	Cell fitness
Functional consequence	Functionally Abnormal	Molecular Mechanism Assessed	DNA and Mismatch repair
Classification		Variant Consequences Detected	Loss of function
Functional score	0.05980	Model System	Immortalized human cells
		Detects Splicing Variants?	No
		Detects NMD Variants?	No
		Number of Variants	17,745
		Clinical Performance	
		OddsPath - Normal	0.043 WQ, STRONG
		OddsPath - Abnormal	04.900 PS, STRONG
		OddsPath-ratio from non-primary research (1 Score A to 4, 10000)	

Score set: MSH2 LOF scores (HAP1)



Case 1: Would you apply PS3_str for this variant now?



Learnings from this case:

- This is not typical MSH2-related Lynch syndrome patient
- Small risk that could be biallelic form (CMMRD) but only 1 case report in literature
- Most likely to be an incidental finding unrelated to clinical presentation
- Patient is age 8 and has a difficult tumour to operate/treat therefore lots of familial anxiety
- Variant is mid-VUS, the functional data could move to likely pathogenic or likely benign depending on result.
- **Functional data is on borderline between PS3 and BS3 and should not be used in variant classification**



Case 2: 56 year old with bowel cancer and IHC showed loss of MSH2/MSH6 IHC



MSH2 c.1276G>C p.(Gly426Arg)

Summary

In Silico Predictions

Control/Case Frequency

Functional Analysis

Click [here](#) to view the current evidence strengths applicable for functional assays, as quantified by CanVIG-UK. Assays with data currently stored in CanVar-UK are listed [here](#).

[Jia et al., 2021](#)

Loss-of-function score	-4.24
Loss-of-function classification	Functional (No LOF)



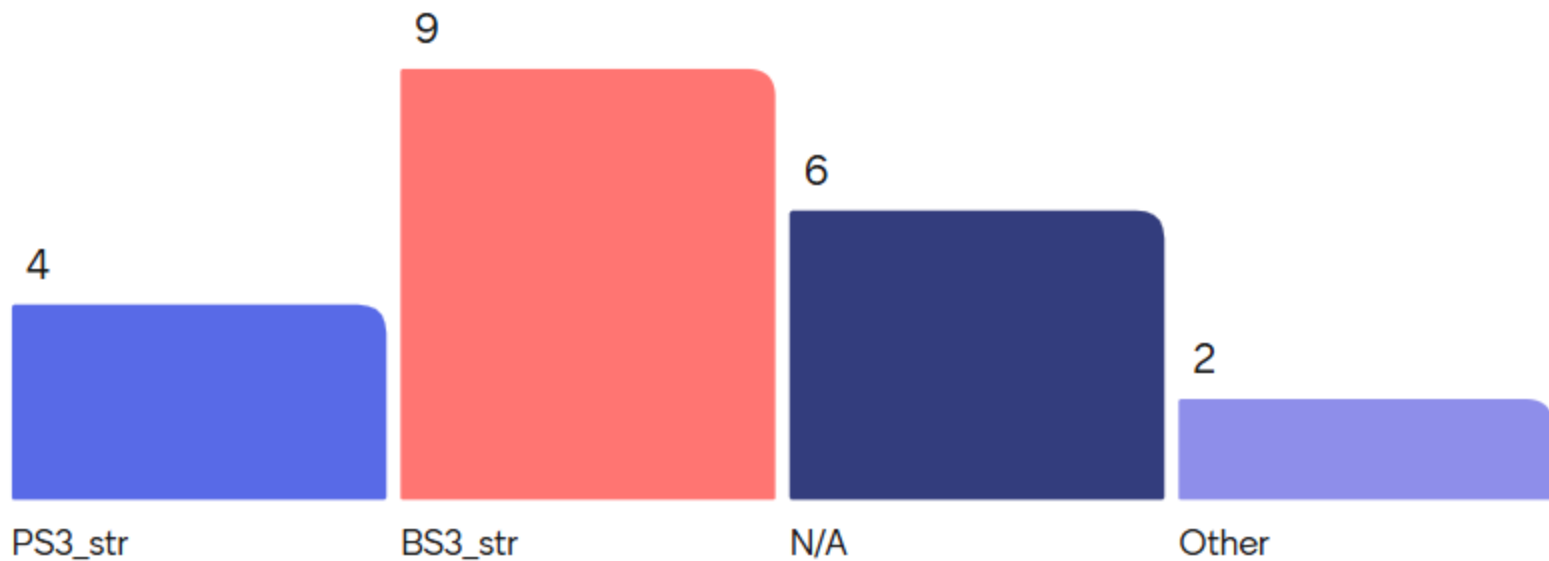
The Programme

CanVIG-UK

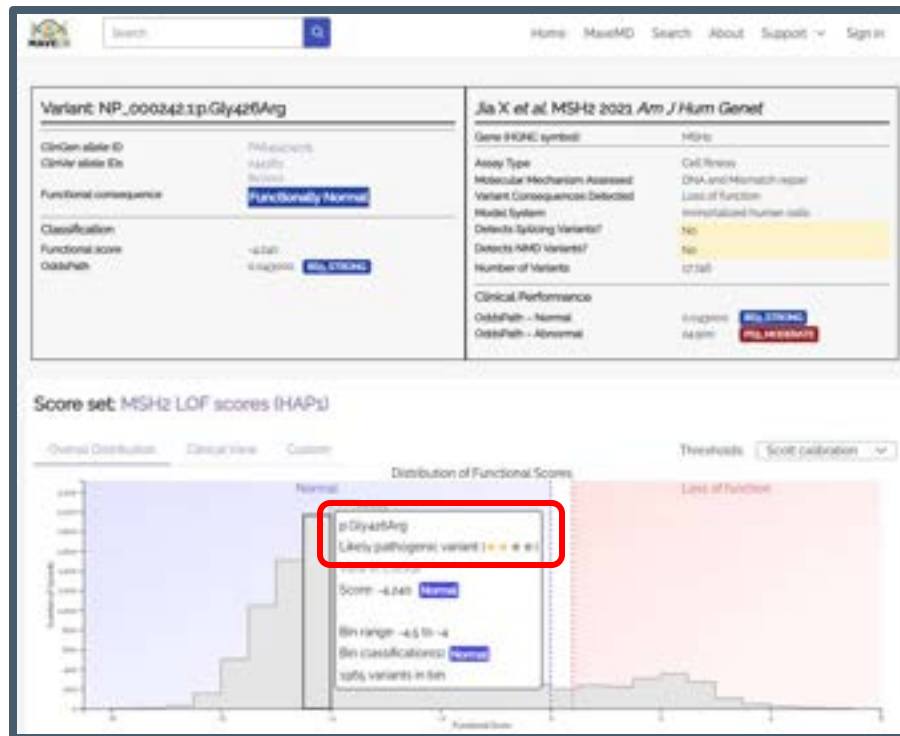
CanVar-UK

1	2	3	Evidence Weighting	
			PS3	BS3
Gene	Reference	PMID		
MSH2	Jia et al, 2021	33357406	Strong	Strong
21				

Case 2: How would you score the functional data?



Case 2



Case 2

NM_000251.3(MSH2):c.1275G>C

Submissions - Germline

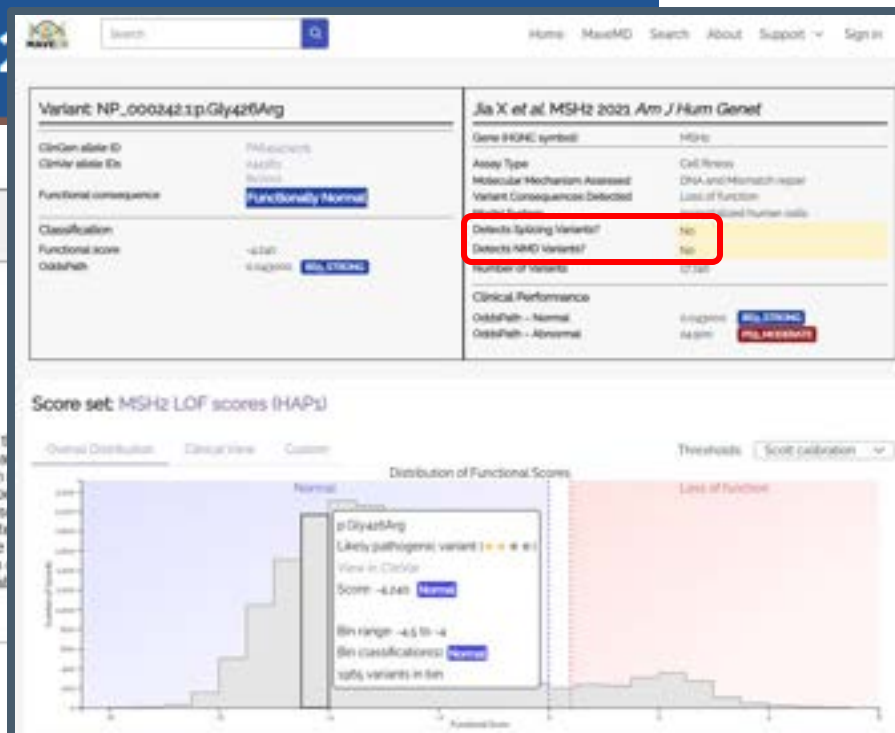
Classification  (Last evaluated)	Review status  (Assertion criteria)	Condition 
Likely pathogenic (Jul 30, 2025)  Contributing to aggregate classification	☆☆☆☆ (Ambry Variant Classification Scheme 2023)	Hereditary cancer- predisposing syndrome

Comment:

The c.1275G>C variant (also known as p.G426R), located in coding exon 7 of the MSH2 gene, results in a change from glycine to arginine at codon 426, an amino acid with dissimilar properties. This alteration is likely to have some effect on normal mRNA splicing. This alteration has been identified in a proband with metachronous CRC at 44 years; this individual had two additional family members with CRC (Cancer 2010 Oct;10:544). This variant has been identified in a proband(s) whose immunohistochemistry (Liu S et al. J. Med. Genet. 2010 Jan;57:62-69; Ambry Int. Aggregation Database (gnomAD). The nucleotide and amino acid positions are altered, which will weaken the native splice donor site and will result in the creation of a cryptic splice site, predicted to be deleterious by in silico analysis. Based on the majority of available evidence, this variant is classified as likely pathogenic.

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown



Learnings from this case:

- Variant affects splicing
- MAVEMD highlights that this assay is not suitable for splicing variants and that variant is LP on ClinVar
- **Ensure assay covers your variant-specific mechanism of disease.**
- **Do not apply this functional data for this variant**



Case 3: 52 yrs female with breast cancer. Family history of BC is all maternal, mother doesn't carry the variant; paternally inherited with no relevant paternal history.

BRCA1: c.301+1G>A p.?

DECIPHER GRCh38 About Browse My Patients DCO (R) Search DECIPHER

BRCA1 17:43104867 C > T

Splice donor variant ENS00000357654 null
 No identical DECIPHER patient variants are accessible to you
 Not in DCO research variants dataset
[Present in ClinVar](#)
 gnomAD - Exome MPF: 4.73×10^{-6}
 Found in 1 cancer proband

Gene: [Browse](#) **Protein** [Annotation](#) [Matching patient variants](#) **30** [Matching DCO research variants](#) **5**

BRCA1: P38398 [Link](#) [Transcripts and More](#)
 Transcript used in protein view: ENS00000357654 v. NM_007294.4 25 exons, 1953aa [MIM: 251250](#)

SpliceAI scores

Variant	Gene	☐ HAVE Select transcript ☐ non-coding transcript	A type	A score	position	REF score	ALT score
NC_000017.10:g.43156884C>T => 17:43156884 C>T	BRCA1 (ENSG00000120438.30/ENST00000105438.14/NM_007294.4) protein-coding (HAV Select transcript (InfoView))	☒ HAVE Select transcript ☐ non-coding transcript	Splice Loss	0.57	89bp	0.77	0.40
splice donor variant	ClinVar , GISA , gnomAD , ClinGen , ExonKit , Outlier , GeneCards		Donor Loss	0.47	1 bp	0.47	0.00
UCSC: gnomAD			Splice Gain	0.00		0.00	0.00
			Donor Gain	0.04	405 bp	0.02	0.07

HAVE Select transcript [All Transcripts](#)

Pangolin scores

Variant	Gene	A type	A score	position	REF score	ALT score
NC_000017.10:g.43156884C>T => 17:43156884 C>T	BRCA1 (ENSG00000120438.30/ENST00000105438.14/NM_007294.4) protein-coding (HAV Select transcript (InfoView))	Splice Loss	0.50	1 bp	0.56	0.00
splice donor variant	ClinVar , GISA , gnomAD , ClinGen , ExonKit , Outlier , GeneCards	Splice Gain	0.02	405 bp	0.00	0.23
UCSC: gnomAD						

Functional data – Findlay Assay 2018 = LOF

Published in final edited form as:

Nature. 2018 October ; 562(7726): 217–222. doi:10.1038/s41586-018-0461-z.

Accurate classification of *BRCA1* variants with saturation genome editing

Gregory M. Findlay¹, Riza M. Daza¹, Beth Martin¹, Melissa D. Zhang¹, Anh P. Leith¹, Molly Gasperini¹, Joseph D. Janizek¹, Xingfan Huang¹, Lea M. Starita^{1,2,*}, and Jay Shendure^{1,2,3,*}

¹Department of Genome Sciences, University of Washington, Seattle, WA 98195

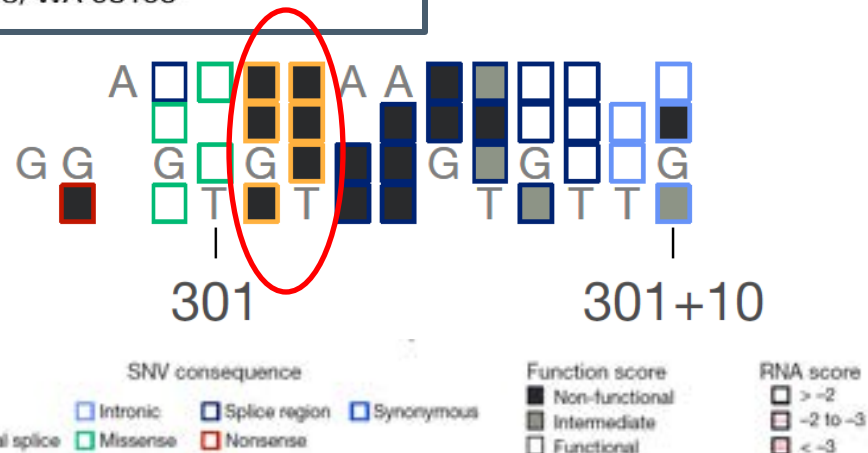
²Brotman Baty Institute for Precision Medicine, Seattle, WA 98195

³Howard Hughes Medical Institute, University of Washington, Seattle, WA 98195

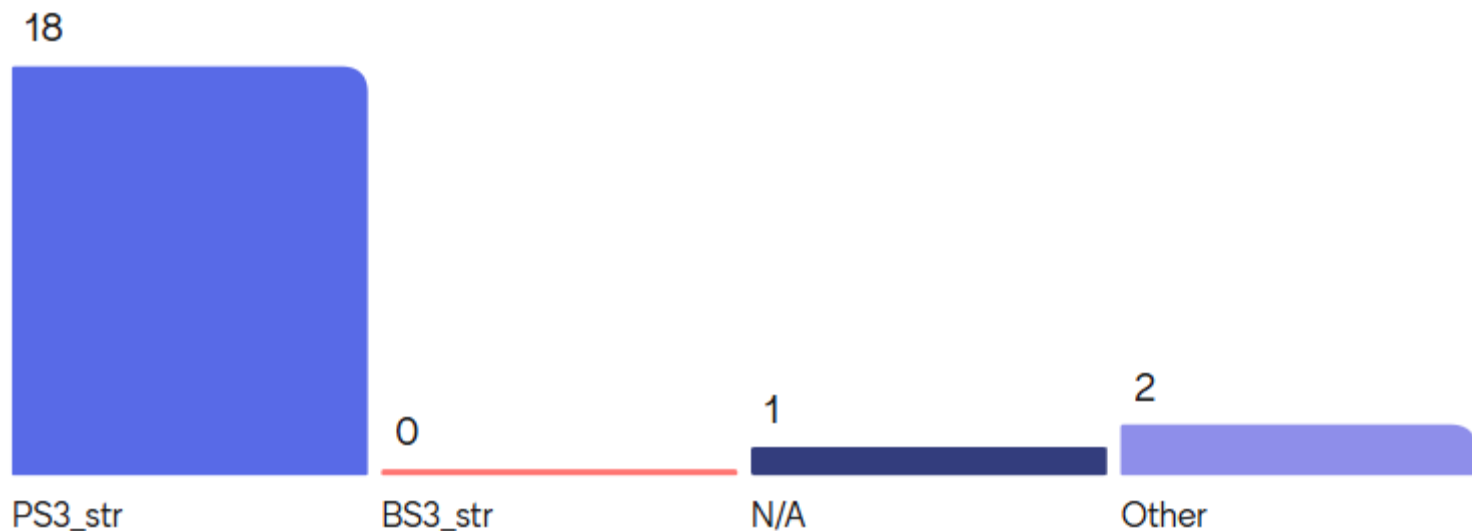
CanVIG-UK Functional Studies Scoring ([live sync](https://live.sync/)) [Access Full Scoring Sheet](#)

Summary of functional studies reviewed and scored by CanVIG-UK in accordance to Clinical Genome Resource (CGR) recommendations (Briach et al. 2020).
Click 'Access Full Scoring Sheet' to see the full summary in Google Sheets.

Gene	Reference	PMID	Evidence Weighting		Additional CanVIG-UK recommendations
			PLS	BLS	
BRCA1	Findlay et al. 2018	30127129	Strong	Strong	MAF functional score < 1.00 survival scores, high MAF



Case 3: How would you score this functional data?

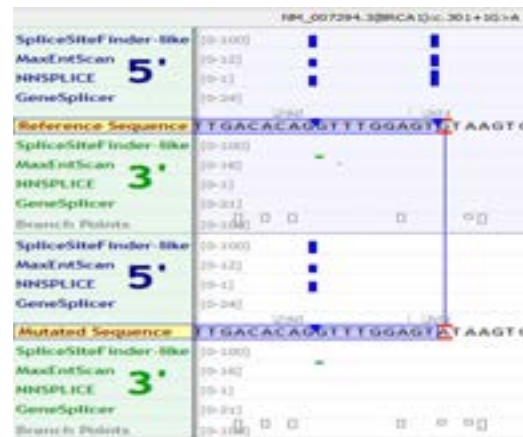


Contradictory evidence = BP2: observed *in trans* with a pathogenic variant in unaffected individual in her 30's

Submitter	Supporting information (See all)
Ambry Genetics Accession: SCV000185671.6 Submitted: (Nov 30, 2020)	Evidence details Publications PubMed (2) Comment: The c.301+1G>A intronic variant results from a G to A substitution one nucleotide after coding exon 4 of the BRCA1 gene. This nucleotide position is highly conserved in available vertebrate species. Using the BDGP and ESEfinder splice site prediction tools, this alteration is predicted to abolish the native splice donor site. Alterations at this donor site are known to produce an in frame transcript impacting four amino acids in the BRCA1 RING domain (Ambry internal data). One functional study found that nucleotide substitutions at this donor site are non-functional in a high-throughput, genome editing haploid cell survival assay (Findlay GM et al. Nature, 2018 10;562:217-222). However this alteration has also been identified in trans with a pathogenic mutation in BRCA1 in an individual without features of Fanconi Anemia (Ambry internal data, personal communication). Since supporting evidence is conflicting at this time, the clinical significance of this alteration remains unclear. (less)

Reply from Ambry - *Thanks for reaching out about this variant. It's a challenge and we've been researching it for a long time. The information we have about co-occurrence comes from collaborating with another lab who tested a shared family member. Our proband had ovarian cancer and was found to carry BRCA1 c.843_846delCTCA. Her daughter was tested at another lab and was found to carry the same pathogenic mutation as her mother as well as BRCA1 c.301+1G>A. She was unaffected in her 30s. It is most likely that these variants are in trans and inherited rather than a de novo in cis event.*

Case 3: *BRCA1* c.301+1G>A



Variant	Gene	Transcript	Alt type	A score	position	REF score	ALT score
NC_000017.10:g.41256884C>T	BRCA1 [170000000:120450_10/100000000:1040134/100000000]	BRCA1-AS1	Accepter Loss	0.37	301bp	0.77	0.40
g.41256884C>T	BRCA1 [170000000:120450_10/100000000:1040134/100000000]	BRCA1-AS1	Donor Loss	0.47	1bp	0.67	0.00
g.41256884C>T	BRCA1 [170000000:120450_10/100000000:1040134/100000000]	BRCA1-AS1	Accepter Gain	0.00	301bp	0.00	0.00
g.41256884C>T	BRCA1 [170000000:120450_10/100000000:1040134/100000000]	BRCA1-AS1	Donor Gain	0.04	301bp	0.00	0.07

Transcript

Variant	Gene	Transcript	Alt type	A score	position	REF score	ALT score
NC_000017.10:g.41256884C>T	BRCA1 [170000000:120450_10/100000000:1040134/100000000]	BRCA1-AS1	Splice Loss	0.30	1bp	0.56	0.00
g.41256884C>T	BRCA1 [170000000:120450_10/100000000:1040134/100000000]	BRCA1-AS1	Splice Gain	0.00	10bp	0.20	0.23

Salisbury accredited NHS lab data: RNA studies confirmed use of cryptic splice donor site at c.293, resulting in r.293_301del p.(Gly98_Tyr101delinsAsp); in-frame deletion of 3 aa in RING domain

CanVIG variant of the month Nov 2021

2. Variants of the month

- a. **BRCA1 c.301+1G>A (Stacey Daniels)**
 - o Pre-meeting classifications: 3/5 labs classified as likely pathogenic, 2/5 as VUS
 - o In meeting poll results (26 responses): 8% likely pathogenic, 69% likely pathogenic with reduced penetrance, 23% VUS
- b. **BRCA1 c.301+1G>T (Megan Carney)**
 - o No pre-meeting classifications
 - o In meeting poll results (26 responses): 19% pathogenic, 62% likely pathogenic with reduced penetrance, 19% VUS
- Main discussion points:
 - o Strategy for dealing with canonical splice site variants flagged by ENIGMA
 - o Application of PM4 vs PVS1 vs PS3 based on RNA assay results in combination with Findlay et al results for c.301+1G>A variant
 - o Possibility of variants at c.301+1 position being hypomorphic based on frequency in population vs Manchester case series and apparent lack of fanconi anaemia phenotype in Ambry case with likely biallelic *BRCA1* variants (further information requested)
 - o Strategy for dealing with apparently conflicting evidence in face of potential hypomorphism

Case 4: c.301+1G>C detected as additional finding in 36yo female 100K (Neur) patient

- May 2022: We reported as likely pathogenic with possible reduced penetrance (PVS1_sup/PS3_str/PM2_mod) and posted on CanVar forum

#65787

Posted by: Miranda Dorkie (ICR) 2022-05-26 14:52:53

Proposed classification: 4 - Likely pathogenic

CAUTION: may be reduced penetrance... ENIGMA guidelines: PVS1_sup (naturally occurring in-frame RNA isoform), LOF on Findlay et al 2016 (-3.59), not on gnomAD, all splice tools (inc. Splice AI) predict loss of donor splice site, RNA studies by Lemari et al 2016 confirmed 9nt (3aa) deletion, X6 ClinVar and X1 LOVD. Found in 100K AF case. We reported as Likely pathogenic (PVS1_sup/PM2_mod/PS3_str) but with possible reduced penetrance based on another variant at this position. Update from clinician: her mother was diagnosed and died of BC aged 42

- Update from clinician – patient's mother died from BC aged 42
- Nov 2022 query from another lab about classification
- My reply = confident in SGE assay
- My musing on alternative splicing and rescue transcripts to CT → email to Greg Findlay

Reply from Greg Findlay...

“The bottom line is I don’t think I can provide a confident answer due to the way this particular exon was assayed in our system...”

“the specific mechanism proposed for functional rescue calls into question whether the SGE result should be given its usual weight...”

“in exon 5 we introduced a synonymous mutation with each variant to be tested, c.297G>A. One such synonymous mutation like this is essential for performing this functional assay with CRISPR, and at least 99% of the time this has no impact on assay result.

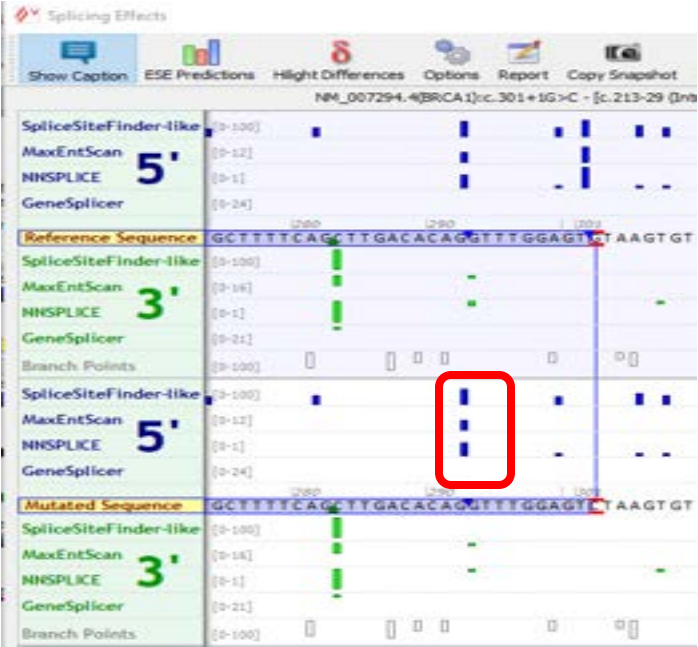
“In this case this second variant (c.297G>A) is 5 nt’s upstream of the canonical donor site. That’s relatively close, but still unlikely to influence how variants impact splicing at the canonical site. However, what worries me is that in the context of the c.301G>C mutation, the alternative site may be viable in patients but not in the assay due to the presence of the c.297G>A on the same allele.”

“As it stands, we should repeat the experiment using a different silent mutation and see if the assay result holds. My instinct is there’s a good chance the assay result may not hold”

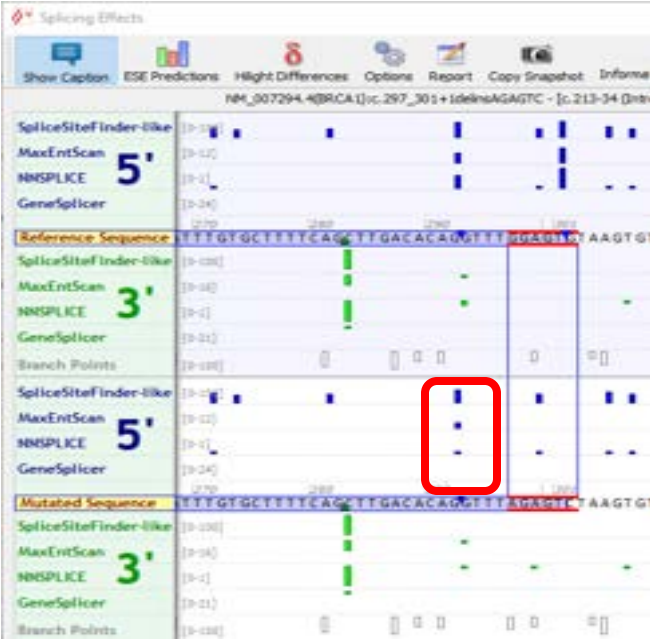


Findlay assay 2018 with c.297G>A “blocking variant” predicted to reduce cryptic splice site too

c.301+1G>C alone



c.301+1G>C with c.297G>A



variant	gene	☒ canonical transcript ☐ non-coding transcript	A type	A score	pre-mRNA position
17-41256889-C-T	BRCA1 [ENSG00000120488.23 / ENSGT00000471381.3 / NM_007300.4]	☒ canonical transcript	Acceptor Loss	0.02	14 bp
UCSC: gnomAD	BRCA1 [ENSG00000120488.23 / ENSGT00000471381.3 / NM_007300.4]	☒ canonical transcript	Donor Loss	0.21	5 bp
	BRCA1 [ENSG00000120488.23 / ENSGT00000471381.3 / NM_007300.4]	☒ canonical transcript	Acceptor Gain	0.00	
	BRCA1 [ENSG00000120488.23 / ENSGT00000471381.3 / NM_007300.4]	☒ canonical transcript	Donor Gain	0.00	

c.297G>A alone on SpliceAI

Updated SGE assay 2023

The Francis Crick Institute
1 Midland Road
London, NW1 1AT, United Kingdom



The Programme

CanVIG-UK

CanVar-UK ¹

National Cancer Genetics MDT

CanVIG-UK Recommendations: BRCA1/BRCA2

v1.30 12 August 2025

CanVIG-UK interim gene-specific recommendations for BRCA1/BRCA2 variants. Additional file: interim report detailing additional testing focused on the 3' end of BRCA1 exon 5, specifically at the c.301+1 position.

Download Guidance

ClinGen VCEP

BRCA1 Exon 5 Memo

Previous Versions

v1.22

v1.21

v1.20

v1.19

v1.18

v1.17

v1.16

v1.15

v1.14

To whom it may concern;

Here, we present a short report entitled, "Evaluation of single nucleotide variants in exon 5 of *BRCA1* using a redesigned Saturation Genome Editing assay".

We have chosen to share the data included here directly with expert clinicians as an interim report in case it may support efforts to classify variants in *BRCA1*. We are taking this unusual step because our results contradict previously published findings (Findlay et al. 2018) regarding the functional consequences of certain variants that have been reported in patients.

Critically, this work has not been peer reviewed. It is part of a larger study on variants in *BRCA1* that is ongoing. All data presented were obtained as part of a research study being conducted at the Francis Crick Institute. As such, we make no claims regarding the data's ultimate validity for adjudicating genetic variants identified in patients. "Function scores" provided here are subject to change at a later date and may be altered in conjunction with a broader analysis that is ongoing.

For further information on the Saturation Genome Editing assay, please see Findlay et al, 2018 (PMID: 30209399).

<https://www.cangene-canvaruk.org/gene-specific-recommendations>

Evaluation of single nucleotide variants in exon 5 of *BRCA1* using a redesigned Saturation Genome Editing assay

Phoebe Dace¹ and Gregory M. Findlay^{1,2}

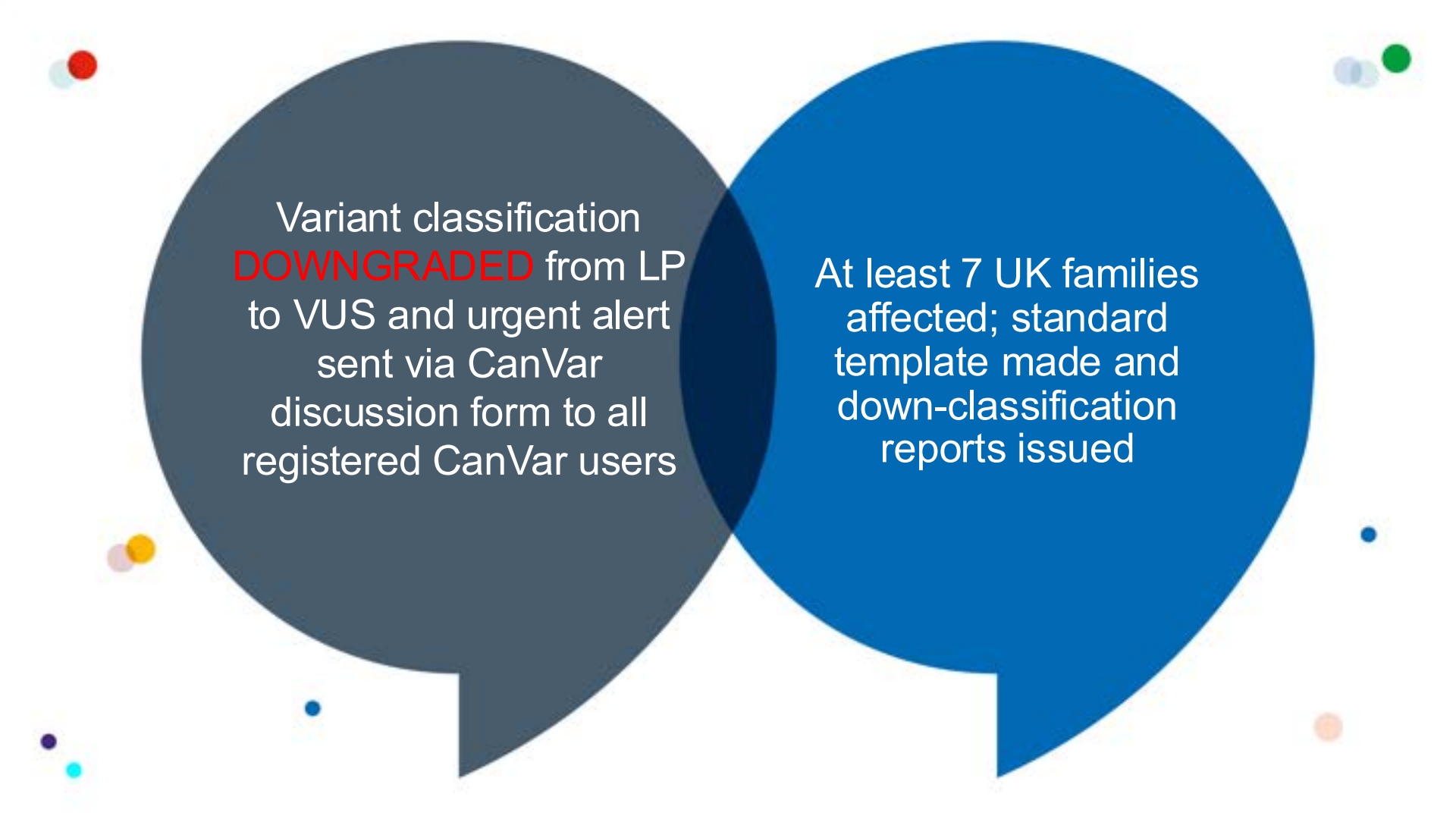
¹ The Genome Function Laboratory, The Francis Crick Institute, London, UK

² Correspondence to greg.findlay@crick.ac.uk

Updated SGE results 2023

Table 1. Variants functionally reclassified using new SGE data for BRCA1 exon 5. All variants assayed in the new SGE experiment that were previously measured to be LoF but are now scored as functional (i.e. neutral) are listed. ClinVar data was obtained on 1 May 2023. Old function scores were obtained from Findlay et al. 2018.

pos hg38	ref	alt	cHGVS	pRGVS	consequence	clinvar	function.score.old	function.score.new	p.value	q.value	old.func.class	new.func.class
43104958	C	G	c.215G>C	p.R71S	NON_SYNONYMOUS	Absent	-2.485	-0.029	0.217	0.482	LOF	FUNC
43104954	C	T	c.215G>A	p.S72N	NON_SYNONYMOUS	Absent	-1.376	-0.073	0.406	0.679	LOF	FUNC
43104938	C	A	c.231G>T	p.T77*	SYNONYMOUS	Likely benign	-1.907	-0.148	0.079	0.250	LOF	FUNC
43104926	T	A	c.243A>T	p.Q81H	NON_SYNONYMOUS	Absent	-1.394	-0.083	0.425	0.693	LOF	FUNC
43104925	G	A	c.244C>T	p.L82F	NON_SYNONYMOUS	Absent	-1.799	-0.188	0.053	0.193	LOF	FUNC
43104919	C	T	c.250G>A	p.E84K	NON_SYNONYMOUS	Uncertain significance	-1.415	-0.022	0.228	0.494	LOF	FUNC
43104918	T	A	c.251A>T	p.E84V	NON_SYNONYMOUS	Absent	-1.578	-0.022	0.228	0.494	LOF	FUNC
43104915	T	C	c.254A>G	p.E85G	NON_SYNONYMOUS	Absent	-1.456	-0.164	0.067	0.232	LOF	FUNC
43104908	C	T	c.261G>A	p.L87*	SYNONYMOUS	Likely benign	-2.272	-0.243	0.028	0.106	LOF	FUNC
43104898	C	A	c.261G>T	p.L87F	NON_SYNONYMOUS	Absent	-2.191	-0.378	0.014	0.087	LOF	FUNC
43104887	C	T	c.301+1G>A	NA	CANONICAL_SPLICE	Conflicting interpretations of pathogenicity	-2.043	-0.093	0.168	0.417	LOF	FUNC
43104887	C	G	c.301+1G>C	NA	CANONICAL_SPLICE	Conflicting interpretations of pathogenicity	-3.592	-0.272	0.792	0.950	LOF	FUNC
43104887	C	A	c.301+1G>T	NA	CANONICAL_SPLICE	Conflicting interpretations of pathogenicity	-2.079	-0.095	0.128	0.349	LOF	FUNC
43104886	A	C	c.301+2T>G	NA	CANONICAL_SPLICE	Absent	-3.765	-0.151	0.076	0.247	LOF	FUNC
43104886	A	G	c.301+2T>C	NA	CANONICAL_SPLICE	Uncertain significance	-2.222	-0.098	0.161	0.406	LOF	FUNC
43104886	A	T	c.301+2T>A	NA	CANONICAL_SPLICE	Absent	-2.424	-0.069	0.160	0.406	LOF	FUNC
43104885	T	A	c.301+3A>T	NA	SPLICE_SITE	Absent	-2.006	-0.072	0.403	0.678	LOF	FUNC
43104885	T	C	c.301+3A>G	NA	SPLICE_SITE	Conflicting interpretations of pathogenicity	-2.165	-0.032	0.212	0.481	LOF	FUNC
43104884	T	A	c.301+4A>T	NA	SPLICE_SITE	Absent	-1.693	-0.002	0.262	0.520	LOF	FUNC
43104884	T	C	c.301+4A>G	NA	SPLICE_SITE	Uncertain significance	-2.487	-0.255	0.024	0.095	LOF	FUNC
43104884	T	G	c.301+4A>C	NA	SPLICE_SITE	Absent	-3.058	-0.262	0.778	0.935	LOF	FUNC
43104883	C	G	c.301+5G>C	NA	SPLICE_SITE	Absent	-2.604	-0.173	0.061	0.216	LOF	FUNC
43104883	C	T	c.301+5G>A	NA	SPLICE_SITE	Uncertain significance	-4.460	-0.045	0.193	0.481	LOF	FUNC
43104882	A	G	c.301+6T>C	NA	SPLICE_SITE	Uncertain significance	-1.949	-0.404	0.935	0.994	LOF	FUNC
43104858	C	G	c.301+10G>C	NA	INTRONIC	Absent	-1.336	-0.526	0.965	0.999	LOF	FUNC



Variant classification
DOWNGRADED from LP
to VUS and urgent alert
sent via CanVar
discussion form to all
registered CanVar users

At least 7 UK families
affected; standard
template made and
down-classification
reports issued

Original data still not updated on other websites ☹️



Variant: NM_007294310.301:G>A		Findlay et al. BRCA1 2018 Nature	
ClinVar community standard allele name NM_007294310.301:G>A		Gene HGNC symbol	BRCA1
ClinVar allele ID	151118	Assay Type	Cell fitness
ClinVar allele ID	151118	Molecular Mechanism Assessed	Molecular function
Functional consequence	Functionally Abnormal	Variant Consequences Detected	Loss of function
Genomic location	chr17:g104685 G104685 chr17:g104685 G104685 chr17:g104685 G104685	Model System	Immunized human cells
Classification		Detects Splicing Variants?	Yes
Functional score	-0.543	Detects NMD Variants?	Yes
ACMG code	PVS1 STRONG	Number of variants	2/93
oddsRatio	52.4	Clinical Performance	
		oddsRatio - Normal	0.000
		oddsRatio - Abnormal	52.400

Score set: BRCA1 SGE Normalized Scores



BRCA Exchange [Detail view](#)

Functional Assay Results

Assays were selected by the ENIGMA consortium as high quality assays that met internal standards permitting estimation of sensitivity and specificity, using approaches as described in [Parsons 2019](#).

- Genetic variation can impact RNA transcripts and/or protein function or abundance. Assays labeled with a **P** measure the impact of genetic variation at the protein level. Assays labeled with a **R/P** measure impact at both the RNA and protein level.
- Reports**, or the authors' interpretation of the functional impact of each variant, are presented as they appear in the publications. A report of 'Many Provided' means that the authors performed more than one assay for each variant but provided no overall report of functional impact.
- Additional assays may be added in the future.

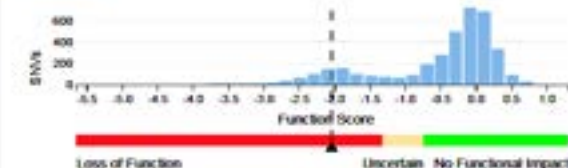
Disclaimer

Functional assay data reflect laboratory models of disease, and should not be used as a substitute for clinical variant interpretation. Further, please note that irrespective of the authors' terms to reflect impact on function, the results presented **do not reflect a final variant classification**.

R/P = Findlay et al, 2018

Report: LQF

This histogram summarizes functional classification results from saturation genome editing, provided by the Starita lab at the University of Washington ([Findlay et al. Nature, 2018](#)). Please see [their site](#) for further information. Please also note unpublished interim report for updated results for variants in Exon 5 from [Dace and Findlay 2023](#).



The median synonymous SVV score: 0.0 and the median non-synonymous SVV score: -0.12

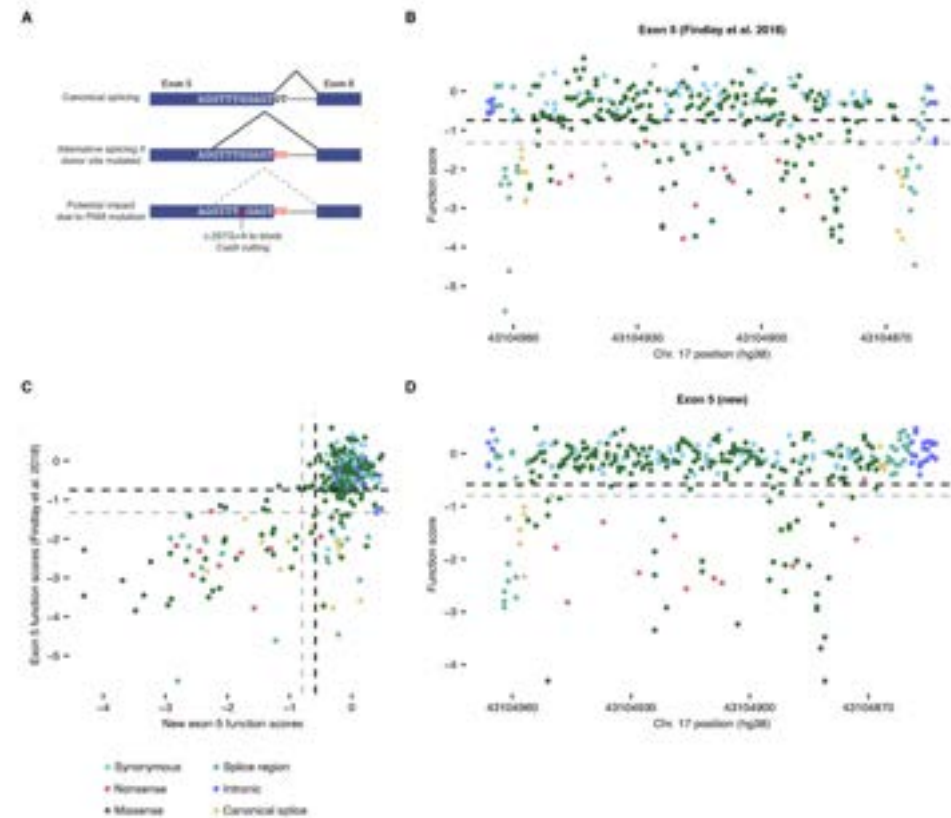
Author: Findlay et al, 2018

New pre-print!!!

Saturation genome editing of *BRCA1* across cell types accurately resolves cancer risk

AUTHORS AND AFFILIATIONS

Phoebe Dace¹, Nicole M. Forrester¹, Maria Zanti², Laura Cubitt¹, Chloé Terwagne¹, Megan Buckley³, Elke M. van Veen³, Thomas Stuart Wilson⁴, Paola Scaffidi^{4,5}, Kyriaki Michailidou² and Gregory M. Findlay¹

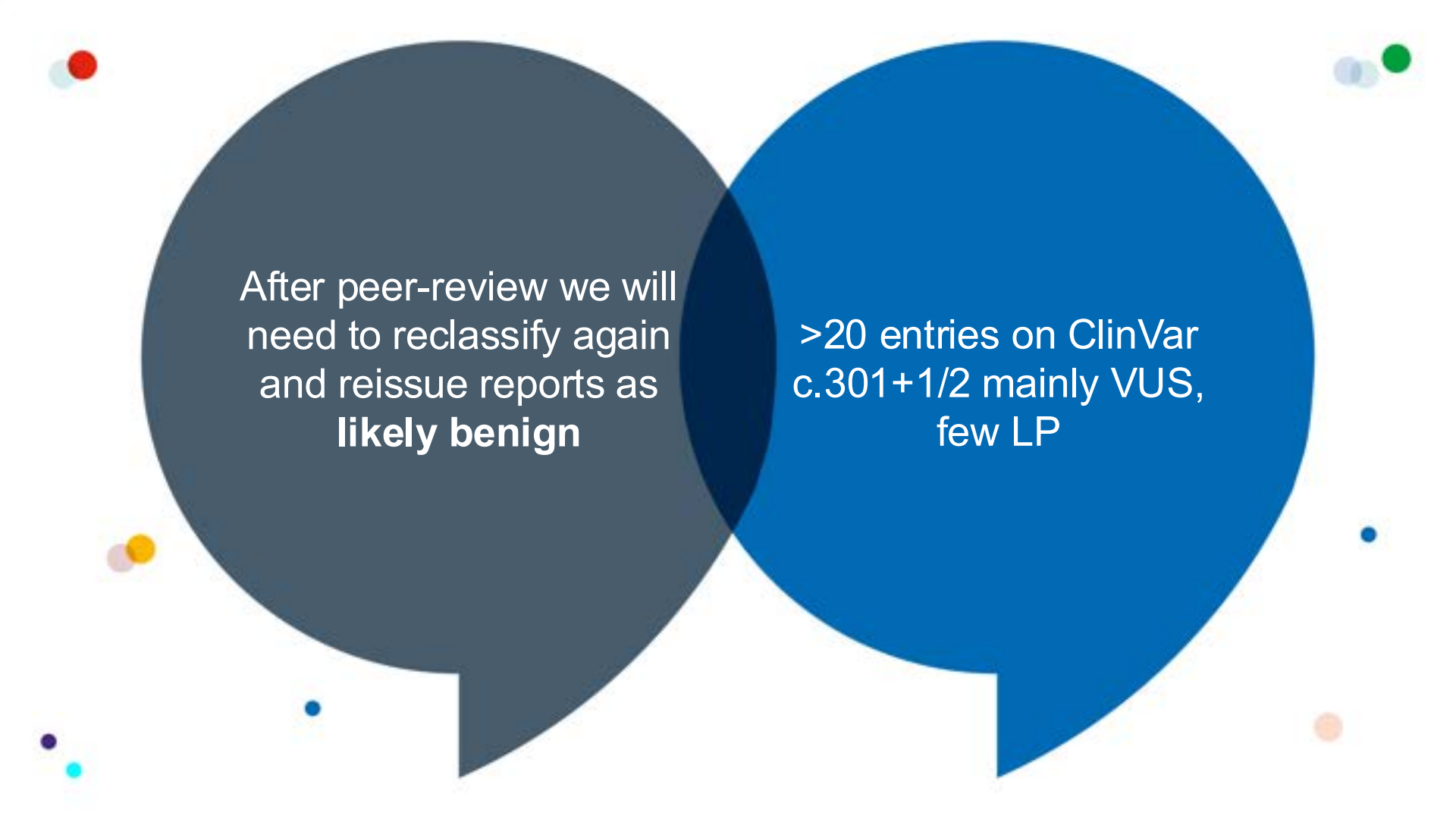


Supplementary Figure 2. A revised SGE library for exon 5 reveals functional rescue of variants impacting the canonical splice donor site.

Learnings from this case:

- Even excellent assays can have errors
- When designing SGE assay ensure synonymous blocking variants do not affect splicing
- **Contact assay authors if there is any conflicting data to ensure it is robust for that variant**
- **Once data is published it is hard to retract/update**





After peer-review we will
need to reclassify again
and reissue reports as
likely benign

>20 entries on ClinVar
c.301+1/2 mainly VUS,
few LP



ClinVar

ClinVar is GREAT for international data sharing and classifications often used as truthsets

However, there are some exceptions/cautions:

Using ClinVar synonymous vs truncating variants:

- Synonymous variants that cause aberrant splicing e.g. *CFTR*: c.1584G>A (p.Glu528=)
- Nonsense variants that are not pathogenic: truncating variants at 5' or 3' ends of genes are not subject to nonsense-mediated decay
 - e.g. truncating variants in exon 9(10) of *BRCA1* (*rescue alternative splicing*); e.g. truncating variants at start or end of genes e.g. *BRCA2*: c.9976A>T (p.Lys3326Ter) (*not subject to NMD*), truncating variants in first 150bp not subject to NMD
- Nonsense variants in last exon can be gain-of-function e.g. *NOTCH2*
- Canonical splice variants that are not pathogenic e.g. *use of cryptic splice site; rescue transcripts*

Using ClinVar LP/P vs LB/B....



ClinVar
Clinically relevant variation

CTGATGGTATGGGGCCAAAGAGATX
AGGTACGGGCTTGTGACTTATGAC
AGGGCTGGGATAAAGTCAAGGGA
CATGGTGCATCTGACTCTCTGAGGA
CAGGTTGGTATCAAGGTTACAAGAI
GGACTGACTCTCTGAGGCTATGG

ClinVar cautions

1. Phenotype data

- Most entries do not provide phenotype and if they do, they don't state whether patient is affected or not
- WES/WGS testing and reporting of all VUS and above findings is common practice in many countries including USA
- Common to have ClinVar entries from individuals with no relevant phenotype
- Problematic where different phenotypes are associated with different mechanisms of disease in same gene



Caution 1: phenotype data

Submissions - Germline 

Classification  (Last evaluated)	Review status  (Assertion criteria)	Condition 	Submitter 	Collapse all rows 	
Uncertain significance (Oct 27, 2017)  Contributing to aggregate classification	☆☆☆☆ (Quest Diagnostics criteria)	not provided	Quest Diagnostics Nichols Institute San Juan Capistrano Accession: SCV000888880.2 First in ClinVar: May 22, 2017 Last updated: Jan 03, 2022	Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: unknown	
Uncertain significance (Apr 13, 2022)  Contributing to aggregate classification	☆☆☆☆ (GeneDx Variant Classification Process June 2021)	Not Provided	GeneDx Accession: SCV000577622.5 First in ClinVar: May 22, 2017 Last updated: Mar 04, 2023	Comment: show Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: yes	
Uncertain Significance (Jan 11, 2024)  Contributing to aggregate classification	☆☆☆☆ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1 (Autosomal dominant inheritance)	All of Us Research Program, National Institutes of Health Accession: SCV004838932.1 First in ClinVar: Apr 20, 2024 Last updated: Apr 20, 2024 Comment: This study involves interpretation of variants in research participants for the purpose of population health screening. Participant phenotype was not available at the time of ... (more)		

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown
Number of individuals with the variant: 1
Zygosity: Single Heterozygote

Example *BRCA1*:
c.301+1G>A
13 submissions
on ClinVar – 12
have no
phenotype data

Likely pathogenic
(Aug 20, 2024)

Contributing to
aggregate classification

☆☆☆
(ACMG Guidelines, 2015)

Breast-ovarian cancer,
familial, susceptibility to, 1
(Autosomal dominant inheritance)

Hereditary Cancer Clinic,
Medical College of Georgia
Accession: SCVD05199846.2
First in ClinVar: Sep 29, 2024
Last updated: Apr 13, 2025

Comment:

PS3 Well-established in vitro or in vivo functional studies supportive of a damaging effect on the gene or gene product. PM1 Located in a mutational hot spot and/or critical and well-established functional domain (e.g., active site of an enzyme) without benign variation. PM4 Protein length changes as a result of in-frame deletions/insertions in a nonrepeat region or stop-loss variants. PP4 Patient's phenotype or family history is highly specific for a disease with a single genetic etiology. BP2 Observed in trans with a pathogenic variant for a fully penetrant dominant gene/disorder or observed in cis with a pathogenic variant in any inheritance pattern. (less)

Observation 1

Collection method: clinical testing

Allele origin: germline

Affected status: yes

Clinical Features:

Carcinoma of fallopian tube (present)

Comment on clinical features:

High-grade serous carcinoma of the fallopian tube

Indication for testing: Ovarian cancer

Zygosity: Single Heterozygote

Age: 60-69 years

Sex: female

Ethnicity/Population group: Caucasian

Comment on evidence:

BRCA1 c.301+1G>A abolishes the intron 5 donor site. RNA analysis of the fallopian cancer in this patient revealed a single alternate transcript (c.293_301del (p.G98_Y101delinsD)) due to the use of an upstream cryptic donor site in exon 5. This in-frame variant replaces GLEY with D in the RING domain. The fallopian cancer harbored a Pathogenic Variant in BRCA1 c.5066T>A (p.Met1689Lys), likely representing the second hit (17q maintained). The tumor scored positive 24% for gLOH at 24% (Caris). An unaffected sister and a daughter are known to carry the variant, AMBRY (RNA, blood sample). The RNA data demonstrated only c.293_301del. No other aberrant transcripts were observed as a result of this alteration. This transcript shows a novel donor gain at c.293, which deletes the final 9 nucleotides of coding exon 4 (total exon 5). The resulting protein impact is p.G98_Y101delinsD. NATERA (RNA, tumor). This reported BRCA1 c.301+1G>A event is a splice donor site variant in the canonical splice site in intron 5, upon reviewing RNA alignments specific to this region: Wild type RNA reads harboring reference allele C demonstrate a preserved, functional splice junction and correct exon 4 and exon 5 splicing is observed by the junction read mapping; RNA reads that support the alternative T allele show loss of splice donor site as those reads with the alternative allele span the exon intron boundary and suggest evidence of dysfunctional splicing mechanism and potential AA codons beyond the splice site in the intronic sequence. There is no evidence of exon 4 skipping. Overall, there is clear evidence of the presence of a wild type BRCA1 transcript and some evidence for an in-frame transcript beyond the impacted exon 4/intron 5 splice donor site. (less)

Platform type: Next-generation sequencing

Testing laboratory: Ambry

Date variant was reported to submitter: 2022-09-19

Testing laboratory interpretation: Uncertain significance

1 entry has
phenotype data
(but is using old
functional data)

Are LOF variants in *COL1A2* disease-causing?

HGNC Approved Gene Symbol: *COL1A2*

Cytogenetic location: *7q21.3* Genomic coordinates (GRCh38) : 7:94,394,895-94,431,227 (from NCBI)

Gene-Phenotype Relationships

Location	Phenotype	View Clinical Synopses	Phenotype MIM number	Inheritance	Phenotype mapping key
<i>7q21.3</i>	{Osteoporosis, postmenopausal}		166710	AD	3
	Combined osteogenesis imperfecta and Ehlers-Danlos syndrome 2		619120	AD	3
	Ehlers-Danlos syndrome, arthrochalasia type, 2		617821	AD	3
	Ehlers-Danlos syndrome, cardiac valvular type		225320	AR	3
	Osteogenesis imperfecta, type II		166210	AD	3
	Osteogenesis imperfecta, type III		259420	AD	3
	Osteogenesis imperfecta, type IV		166220	AD	3



Evaluation of *COL1A2* LOF curations for OI

Decipher – states that LOF are pathogenic in heterozygous state, however the publication they reference is a frameshift that escapes NMD. Communication from Decipher – curation is not manual and is not intended to be definitive, but is a prediction based on PMID 39172982 (Proteome-scale prediction of molecular mechanisms underlying dominant genetic diseases) supplemented by literature hits.

ClinGen – state *COL1A2* is haplosufficient.

Skeletal dysplasias VCEP – state *COL1A2* is haplosufficient.

GeneReviews – does not cover this topic.

HGMD – Most of the null variants curated as ‘DM’ are in association with non-OI phenotypes. **Only 3 papers presenting null variants as causative of OI (authors appear to have assumed LOF is a mechanism of disease).**

ClinVar – 35 null variants reported mainly LP/P. Most do not have phenotypes reported. Labs likely to submit the highest possible classification to ClinVar (these variants are pathogenic for very rare AR cvEDS) → circularity bias

CVA – 7 individuals with null variants. None have any symptoms of OI reported.

GnomAD – v4 contains 95 individuals with heterozygous null variants. This is not informative (reduced penetrance, phenotypic variability, delayed onset).

UKBB – 31 cases – none with OI; BMD not significant; fractures in last 5 years not significant

Sheffield cases – SDGS has ran the specialist service for OI in England for 20 years. Null variants in *COL1A2* are a very rare occurrence, audit underway, so far only found once in a patient with OI symptoms.



ClinVar cautions

2. 1 or 2 star classifications

- May be based on data from few labs; information propagation bias; data circularity

Review status

- ☐ Practice guideline (637)
- ☐ Expert panel (11,522)
- ☐ Multiple submitters (367,258)
- ☐ Single submitter (1,835,121)
- ☐ At least one star (2,307,284)
- ☐ Conflicting classifications (93,028)



Germline classification

- ☐ Conflicting classifications (93,395)
- ☐ Benign (136,188)
- ☐ Likely benign (570,492)
- ☐ Uncertain significance (1,458,519)
- ☐ Likely pathogenic (66,466)
- ☐ Pathogenic (104,058)

Types of conflicts

- ☐ P/LP vs LB/B (660)
- ☐ P/LP vs VUS (14,006)
- ☐ VUS vs LB/B (79,558)

ClinVar cautions

3. Variant classifications guidelines used

- Often not ACMG or own versions e.g. Sherlock, or over-use disallowed codes (PP5)
- Some entries are old before multiple ClinGen/ACMG updates

Likely pathogenic
(Feb 15, 2024)
☆☆☆
(ACMG Guidelines, 2015)
C Contributing to
aggregate classification

Pathogenic
(Jan 26, 2025)
☆☆☆
(Invitae Variant Classification
Sherlock (09022015))
C Contributing to
aggregate classification

Pathogenic
(Jan 09, 2025)
☆☆☆
(GeneDx Variant Classification
Process June 2021)
C Contributing to
aggregate classification

Uncertain
significance
(Jul 30, 2020)
☆☆☆
(Ambry Variant Classification
Scheme 2023)
C Contributing to
aggregate classification

Likely pathogenic
(Jan 24, 2019)
☆☆☆
(LabCorp Variant Classification
Summary - May 2015)
C Contributing to
aggregate classification



ClinVar cautions

4. Erroneous classifications

- New data supercedes previous classification e.g, gnomAD v4.1



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NM_000089.4(COL1A2):c.2827G>A (p.Gly943Arg)



We've updated the full submission template; previous versions are no longer supported
[more about the available templates.](#)

Germline

Classification



Conflicting classifications of pathogenicity

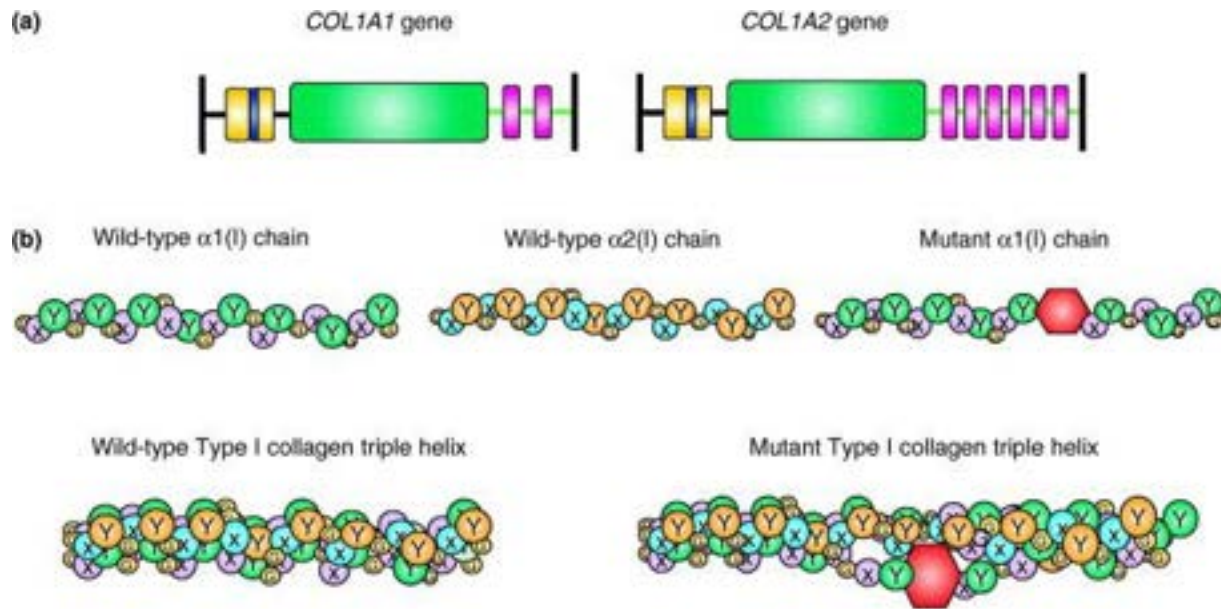
Pathogenic(2); Likely pathogenic(2); Uncertain significance(1)

5 out of 5 submissions contributed to this classification

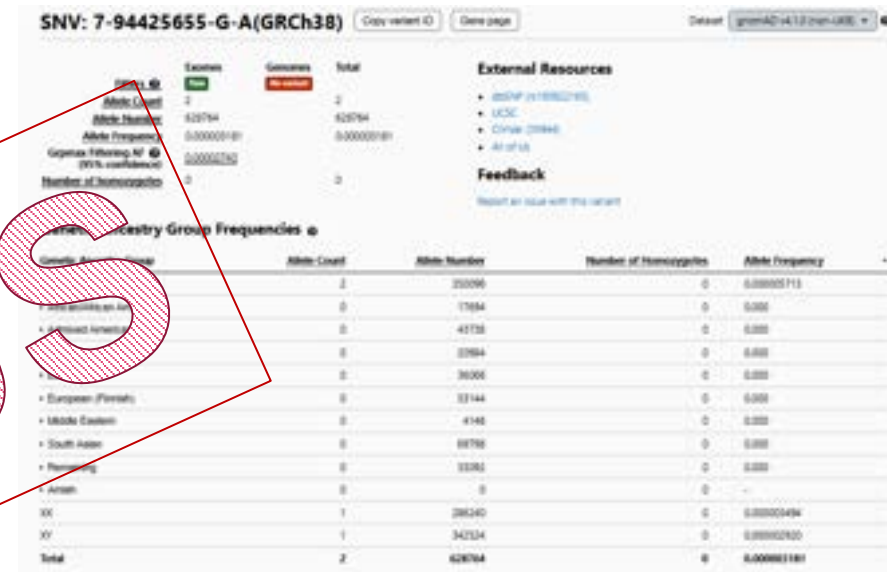
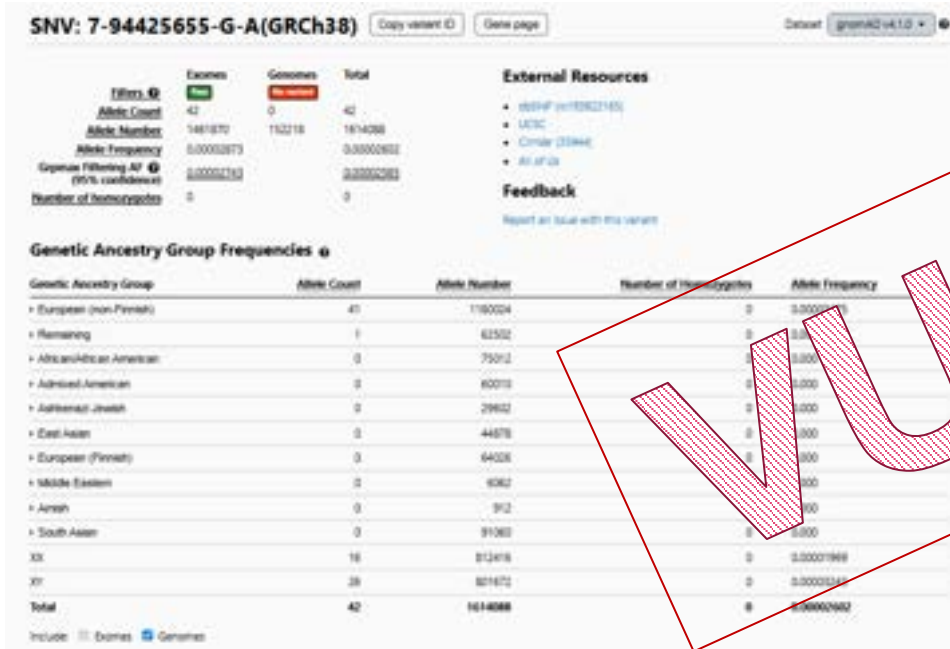
Evidence of favour of pathogenicity

- REVEL 0.978; CADD 33
- Glycine altering – most pathogenic variants are Glycines disrupting the Gly-X-Y repeat critical to triple-helix
- 4/5 ClinVar submissions are P or LP (last 2 are P)
- p.(Gly943Glu) reported in patient with OI (Bardai (2016) PubMed: 27509835)
- Seen in patients with OI (ClinVar)
- Not in gnomAD v2.1 or v3.1

ACMG: PM1_str/PP3_sup/PM2_sup/PM5_sup/PS4_mod = LP
with 9 points



New evidence....



Phenotypic data from UKBB: 0/44 have ICD10 for OI (Q78.0);
5/44 (11.3%) had fracture in last 5 years vs 9.6% whole UKB (not significant).
Heel BMD available for 20/44 and mean is 0.547, median is 0.53g/cm2 (whole UKB median 0.53)

Contacted all ClinVar labs but they will not amend their classification unless data is published ☹️

Solutions?

- Remove/resolve outliers?
- Verify data using clinically validated datasets from affected patients?
- Caution if you have no clinical data for pathogenicity, only predictive/functional data!





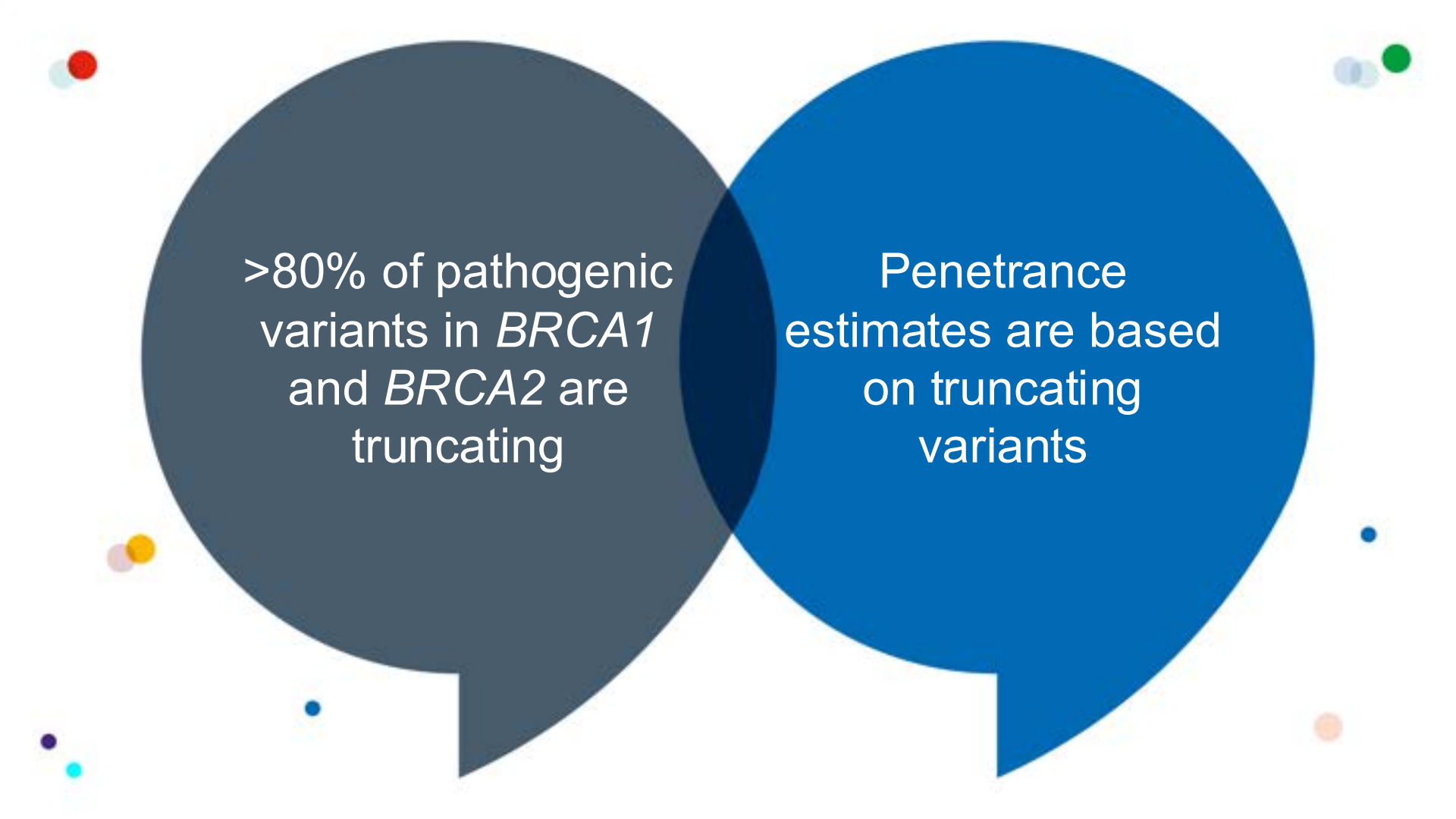
Penetrance

Penetrance *BRCA1* and *BRCA2*

Female <i>BRCA1</i> Germline Pathogenic Variant (GPV) carrier age-dependent cumulative cancer risks (95% confidence interval)			Female <i>BRCA2</i> Pathogenic Variant (GPV) carrier age-dependent cumulative cancer risks (95% confidence intervals) *		
Age (years)	Breast cancer (Note population risk to age 80 approx 10%*)	Tubo-ovarian cancer (Note population risk to age 80 approx 1.5%*)	Age (years)	Breast cancer (Note population risk to age 80 approx 10%*)	Tubo-ovarian cancer (Note population risk to age 80 approx 1.5%*)
21-30	4% (2-7%)	-	21-30	4% (2-9%)	-
31-40	24% (21-29%)	2% (1-3%)	31-40	13% (9-19%)	<0.5%
41-50	43% (39-48%)	8% (6-12%)	41-50	35% (29-41%)	1-2%
51-60	56% (51-61%)	20% (16-26%)	51-60	53% (46-59%)	7% (4-11%)
61-70	66% (61-72%)	41% (33-50%)	61-70	61% (55-68%)	15% (10-23%)
71-80	72% (65-79%)	44% (36-53%)	71-80	69% (61-77%)	17% (11-25%)†

Other cancers	<i>BRCA1</i> GPV carrier risk (95% confidence interval)	Population risk*	Other cancers	<i>BRCA2</i> GPV carrier risk (95% confidence interval)	Population risk*
Male breast cancer	0.4% (0.1-1.5%) by age 80	Rare	Male breast cancer	4% (2-8%) by age 80	Rare
Prostate cancer *Adjusted risk for screening effect, please see FAQ for unadjusted figures	17% (8-26%) by age 85† (Nyberg et al, 2020) No significant association (Li et al, 2022)	Approx 12% by age 80	Prostate cancer *Adjusted risk for screening effect, please see FAQ for unadjusted figures	41% (22-54%) by age 85† (Nyberg et al, 2020) 27% (21-35%) by age 80 (Li et al, 2022)	Approx 12% by age 80
Pancreatic cancer*	Male: Approx 3% by age 80	Approx 1% by age 80	Pancreatic cancer*	Male: Approx 4-5% by age 80	Approx 1% by age 80
	Female: Similar to population risk	Up to 1% by age 80		Female: Approx 2% by age 80	Up to 1% by age 80
No significant association consistently found for other cancers.			No significant association consistently found for other cancers.		

Risk of OC >70 is likely underestimated by referenced study, see management below and FAQ



>80% of pathogenic
variants in *BRCA1*
and *BRCA2* are
truncating

Penetrance
estimates are based
on truncating
variants

Lower penetrance for missense variants

Genetics in Medicine (2022) 24, 119–129



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Medicine

www.journals.elsevier.com/genetics-in-medicine

ARTICLE

Risks of breast and ovarian cancer for women harboring pathogenic missense variants in *BRCA1* and *BRCA2* compared with those harboring protein truncating variants



Li et al PMID: 34906479

Dorling et al. *Genome Medicine* (2022) 14:51
<https://doi.org/10.1186/s13073-022-01052-8>

Genome Medicine

RESEARCH **Open Access**

Breast cancer risks associated with missense variants in breast cancer susceptibility genes



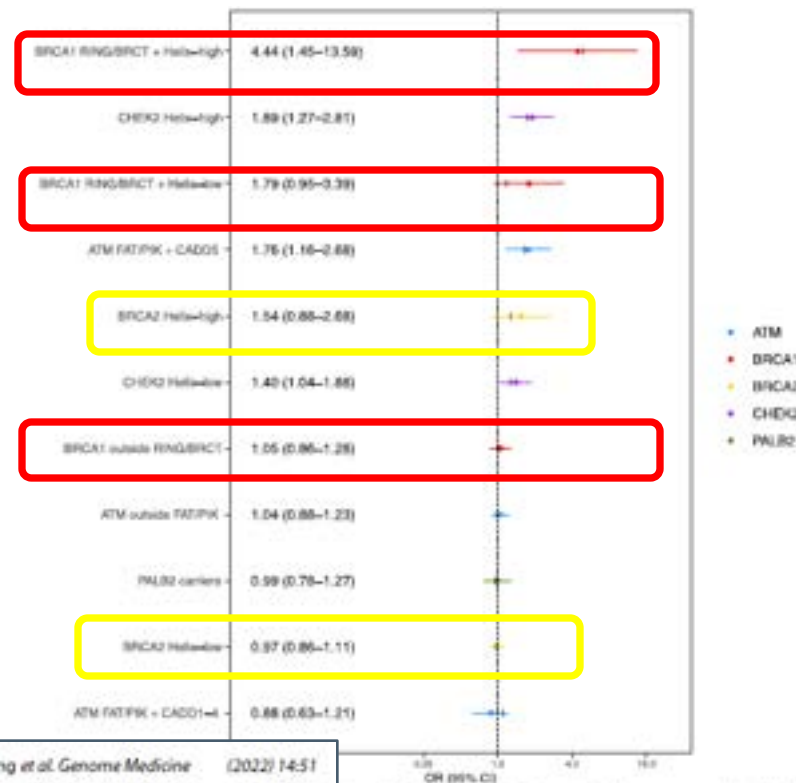
Misense risks vary depending on size of change & location

Table 2 HR estimates by variant/group of variants

Variant/Variant Group	Breast Cancer Dx <50 HR (95% CI)	Breast Cancer Dx ≥50 HR (95% CI)	Ovarian Cancer HR (95% CI)
BRCA1			
GC-HBOC BRCA1 PTC	17.0 (9.4-30.5)	12.8 (5.7-28.7)	27.1 (8.6-85.2)
RING domain			
p.[Cys61Gly]	14.6 (8.8-24.3)	7.6 (3.6-16.4)	41.2 (19.0-89.3)
p.[Cys64Arg]	3.7 (1.4-9.5) ^a	4.1 (0.9-18.3)	99.9 (37.8-264)
Other RING domain	12.0 (3.2-37.4)	2.9 (0.5-17.0)	23.9 (4.6-119)
All RING Domain	11.5 (7.2-18.3)	5.8 (3.0-11.3)	41.0 (20.9-80.4)
GC-HBOC p.[Cys61Gly]	15.3 (8.3-28.2)	3.7 (1.5-9.4) ^a	27.1 (7.2-102.8)
BRCT domain			
p.[Arg369Trp]	10.4 (2.7-39.4)	2.0 (0.2-21.8)	31.5 (5.1-195)
p.[Ala170Glu]	12.1 (4.3-34.0)	4.9 (1.3-18.3)	5.2 (0.8-33.8)
p.[Gly1738Arg]	22.8 (9.2-56.9)	3.6 (1.4-8.9)	14.7 (3.8-57.8)
Other BRCT domain	15.0 (6.0-37.5)	2.1 (0.7-6.2) ^b	18.8 (6.3-56.2)
All BRCT domain	14.8 (8.7-25.1)	2.8 (1.4-5.6) ^b	15.2 (7.6-30.4)
All BRCA1 missense	13.1 (9.2-18.9)	3.9 (2.4-6.2) ^b	21.7 (12.4-38.0)
BRCA2			
GC-HBOC BRCA2 PTC	10.4 (5.9-19.8)	7.0 (3.3-14.7)	13.1 (3.7-45.6)
DBD			
p.[Trp262Cys]	5.0 (0.3-92.0)	1.7 (0.4-6.9)	2.1 (0.1-32.8)
p.[Asp2723His]	8.5 (2.5-28.7)	5.2 (1.5-18.6)	15.0 (2.1-109)
p.[Arg3052Trp]	8.1 (0.9-73.3)	3.5 (0.4-30.0)	4.1 (0.3-115)
Other DBD	9.5 (3.8-23.7)	5.3 (2.1-13.2)	5.6 (1.4-21.8)
All DBD	8.3 (2.2-30.8)	3.9 (2.0-7.2)	5.5 (2.0-14.8)

BRCT, BRCA1 C-Terminal; CI, confidence interval; DBD, DNA binding domain; Dx, Age at diagnosis of breast cancer; GC-HBOC, German Consortium for Hereditary Breast and Ovarian Cancer; HR, hazard ratio; PTC, protein truncating.

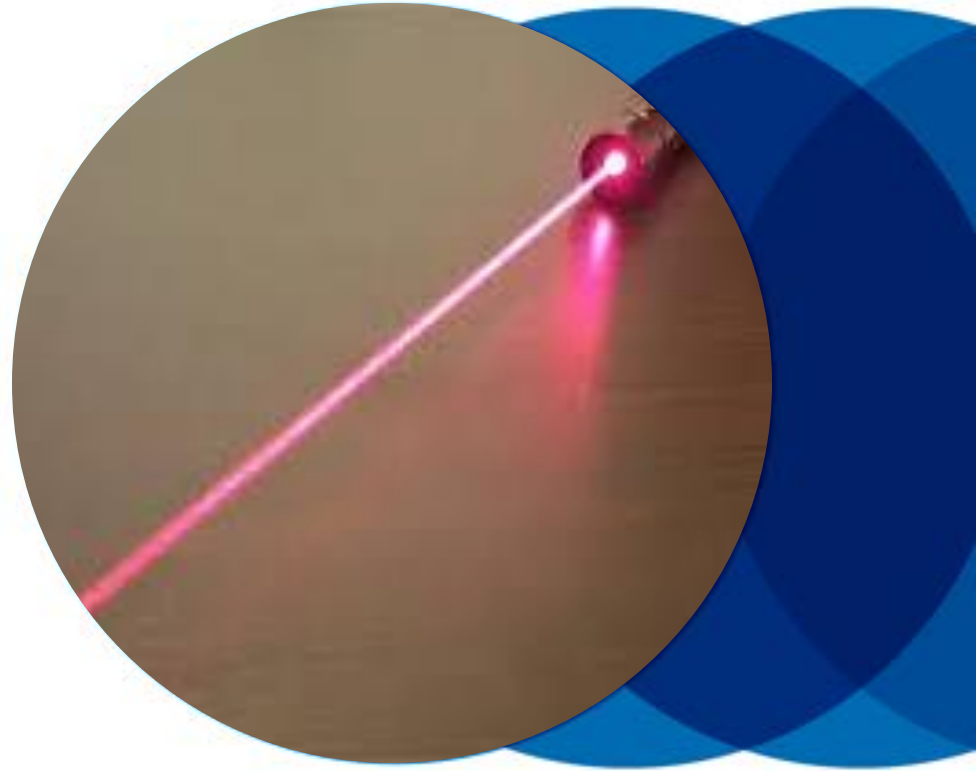
Li et al 2022
GIM



Dorling et al Genome Medicine (2022) 14:51

Fig. 3 Breast cancer risk estimates from composite gene model in validation samples. Black marks indicate corresponding ORs from training

Let's focus only on
clinically proven, known
reduced penetrant *BRCA1*
and *BRCA2* variants...



BRCA1: c.5096G>A p.(Arg1699Gln) – the reduced penetrance poster child

Published in final edited form as:

J Med Genet. 2012 August ; 49(8) : . doi:10.1136/jmedgenet-2012-101037.

BRCA1 R1699Q variant displaying ambiguous functional abrogation confers intermediate breast and ovarian cancer risk

Amanda B Spurdle^{1,2}, Phillip J Whaley¹, Bryony Thompson^{1,2}, Bingjian Feng³, Sue Healey¹, Melissa A Brown⁴, Christopher Pettigrew⁴, kConFab⁵, Christi J Van Asperen⁶, Margreet G E M Ausems⁷, Anna A Kattentidt-Mouravieva⁸, Ans M W van den Ouweland⁹, Dutch Belgium UV Consortium^{9,10}, Annika Lindblom¹¹, Maritta H Pigg¹², Rita K Schmutzler¹³, Christoph Engel¹⁴, Alfons Meindl¹⁵, German Consortium of Hereditary Breast and Ovarian Cancer^{9,13}, Sandrine Caputo¹⁶, Olga M Sinilnikova^{17,18}, Rosette Lidereau¹⁶, French COVAR group collaborators¹⁹, Fergus J Couch²⁰, Lucia Guidugli²⁰, Thomas van Overeem Hansen²¹, Mads Thomassen²², Diana M Eccles²³, Kathy Tucker²⁴, Javier Benítez²⁵, Susan M Domchek²⁶, Amanda E Toland²⁷, Elizabeth J Van Rensburg²⁸, Barbara Wappenschmidt¹³, Åke Borg²⁹, Maaike P G Vreeswijk³⁰, and David E Goldgar^{3,31,9} on

Results—Comparison of *BRCA1* carrier prediction scores of probands using the BOADICEA risk prediction tool revealed that *BRCA1* c.5096G>A p.Arg1699Gln variant carriers had family histories that were less ‘*BRCA1*-like’ than *BRCA1* c.5095C>T p.Arg1699Trp mutation carriers ($p=0.0001$), but more ‘*BRCA1*-like’ than *BRCA1* X family carriers ($p=0.0001$). Further, modified segregation analysis of the subset of 30 families with additional genotyping showed that *BRCA1* c.5096G>A p.Arg1699Gln had reduced penetrance compared with the average truncating *BRCA1* mutation penetrance ($p=0.0002$), with estimated cumulative risks to age 70 of breast or ovarian cancer of 24%.

Conclusions—Our results provide substantial evidence that the *BRCA1* c.5096G>A p.Arg1699Gln (R1699Q) variant, demonstrating ambiguous functional deficiency across multiple assays, is associated with intermediate risk of breast and ovarian cancer, highlighting challenges for risk modelling and clinical management of patients of this and other potential moderate-risk variants.



Cancer genetics
Short Report

The *BRCA1* c. 5096G>A p.Arg1699Gln (R1699Q) intermediate risk variant: breast and ovarian cancer risk estimation and recommendations for clinical management from the ENIGMA consortium

Setareh Moghadasi¹, Huong D Meeks², Maaike PG Vreeswijk³, Linda AM Janssen¹, Åke Borg⁴, Hans Ehrencrantz⁵⁻⁸, Ylva Paulsson-Karlsson⁷, Barbara Wappenschmidt⁹⁻¹⁰, Christoph Engel¹¹⁻¹², Andrea Gehrig¹³⁻¹⁴, Norbert Arnold¹⁵, Thomas Van Overeem Hansen¹⁷⁻¹⁸, Mads Thomassen¹⁹, Uffe Birk Jensen²⁰, Torben A Kruse¹⁹, Bent Ejertsen¹⁸⁻²¹, Anne-Marie Gerdes¹⁹⁻²², Inge Sekkle Pedersen²³⁻²⁴, Sandrine M Caputo²⁵, Fergus Couch²⁶, Emily J Hallberg²⁷, Ans MW van den Ouweland²⁸, Margriet J Collee²⁹, Erik Teugels²⁹, Muriel A Adank³⁰, Rob B van der Luijt³¹⁻³², Arjen R Mersinkamp³³, Jan C Oosterwijk³⁴, Marinus J Blok³⁵, Nicolas Jasin³⁶, Kathleen BM Claes³⁷, Kathy Tucker³⁸, Valeria Vassolo³⁹⁻⁴⁰, Amanda Ewart Toland⁴¹, Diana E Eccles⁴², Peter Devilee³, Christie J Van Asperen¹, Amanda B Spurdle⁴³, David E Goldgar⁴⁴, Encarna Gómez García¹⁵

Results In this cohort the cumulative risk of BC and OC by age 70 years was 20% and 6%, respectively. The relative risk for developing cancer was highest when using a model that included the effects of both the R1699Q variant and a residual polygenic component compared with monogenic model (for BC 3.67 vs 2.83, and for OC 6.41 vs 5.83).

Conclusion Our results confirm that *BRCA1* R1699Q confers an intermediate risk for BC and OC. Breast surveillance for female carriers based on mammogram annually from age 40 is advised. Bilateral salpingo-oophorectomy should be considered based on family history.

Penetrance *BRCA1* and *BRCA2*

Female <i>BRCA1</i> Germline Pathogenic Variant (GPV) carrier age-dependent cumulative cancer risks (95% confidence interval)			Female <i>BRCA2</i> Pathogenic Variant (GPV) carrier age-dependent cumulative cancer risks (95% confidence intervals) *		
Age (years)	Breast cancer (Note population risk to age 80 approx 10%*)	Tubo-ovarian cancer (Note population risk to age 80 approx 1.5%*)	Age (years)	Breast cancer (Note population risk to age 80 approx 10%*)	Tubo-ovarian cancer (Note population risk to age 80 approx 1.5%*)
21-30	4% (2-7%)	-	21-30	4% (2-9%)	-
31-40	24% (21-29%)	2% (1-3%)	31-40	13% (9-19%)	<0.5%
41-50	43% (39-48%)	8% (6-12%)	41-50	35% (29-41%)	1-2%
51-60	56% (51-61%)	20% (16-26%)	51-60	53% (46-59%)	7% (4-11%)
61-70	66% (61-72%)	41% (33-50%)	61-70	61% (55-68%)	15% (10-23%)
71-80	72% (65-79%)	44% (36-53%)	71-80	69% (61-77%)	17% (11-25%)†

Other cancers	<i>BRCA1</i> GPV carrier risk (95% confidence interval)	Population risk*	Other cancers	<i>BRCA2</i> GPV carrier risk (95% confidence interval)	Population risk*
Male breast cancer	0.4% (0.1-1.5%) by age 80	Rare	Male breast cancer	4% (2-8%) by age 80	Rare
Prostate cancer *Adjusted risk for screening effect, please see FAQ for unadjusted figures	17% (8-26%) by age 85 [†] (Nyberg et al, 2020)	Approx 12% by age 80	Prostate cancer *Adjusted risk for screening effect, please see FAQ for unadjusted figures	41% (22-54%) by age 85 [†] (Nyberg et al, 2020)	Approx 12% by age 80
	No significant association (Li et al, 2022)			27% (21-35%) by age 80 (Li et al, 2022)	
Pancreatic cancer*	Male: Approx 3% by age 80	Approx 1% by age 80	Pancreatic cancer*	Male: Approx 4-5% by age 80	Approx 1% by age 80
	Female: Similar to population risk	Up to 1% by age 80		Female: Approx 2% by age 80	Up to 1% by age 80
No significant association consistently found for other cancers.			No significant association consistently found for other cancers.		

Risk of OC >70 is likely underestimated by referenced study, see management below and FAQ



Search MaveDB for human gene variants

 ▼ ▼ ▼

Click here to return to HGVS search:

NM_007294.4(BRCA1):c.5096G>A (p.Arg1699Gln)

ClinGen Allele ID: <http://reg.genome.network/allele/CA003235>

GRCh38 coordinates: NC_000017.11:g.43063930C>T

GRCh37 coordinates: NC_000017.10:g.41215947C>T



 No score sets containing this variant were found in MaveDB. 

BRCA2: c.8351G>A p.(Arg2784Gln)

Predictive test requested on 28yo female
with FHx BC; proband tested in Brazil
and reported as LP

REVEL 0.77/BayesDel 0.40 (PP3)

GnomAD v4.1 – 27 alleles

Conflicting data on ClinVar
and Genetic epidemiology
data.

ENIGMA multifactorial data
= benign (BP5_str)

ClinVar Genomic variation as it relates to human health

Search by gene
Advanced

About Access Submit Stats FTP Help

NM_000059.4(BRCA2):c.8351G>A (p.Arg2784Gln)

 We've updated the full submission template; previous versions are no longer supported. [more about the available templates.](#)

Classification
★☆☆☆

Conflicting classifications of pathogenicity
Pathogenic(1); Likely pathogenic(6); Uncertain significance(9)
16 out of 23 submissions contributed to this classification



BRCA2 c.8351G>A p.(Arg2784Gln)			Download
Summary	In-Silico Predictions	Control/Case Frequency	Genetic Epidemiology
Genetic Epidemiology			
Parsons et al., Human Mutation, 2013			
Combined (Parsons LRIACMG LUR)		0.73e-06 / -15.9	
Segregation (Parsons LRIACMG LUR)		0.000107 / -12.5	
Pathology (Parsons LRIACMG LUR)		1.13 / 0.173	
Co-occurrence (Parsons LRIACMG LUR)		0.0498 / -4.13	
Family history (Parsons LRIACMG LUR)		1.48 / 0.530	
Case-control (Parsons LRIACMG LUR)		1 / 8	
Li et al., Genet Med, 2020			
Likelihood Ratio (Li LRIACMG LUR)		0.89 / 3.11	

BRCA2 c.8351G>A p.(Arg2784Gln)

3 (old) functional assays
showed deleterious
impact on function
= PS3_sup

BRCA2 c.8351G>A p.(Arg2784Gln)	
Summary	In Silico Predictions
Control Case Frequency	
Functional Analysis	
Click here to view the current evidence strengths applicable for functional assays, as quantified by ClinVar-UK. Assays with data currently stored in ClinVar-UK are listed here .	
Hersant et al., 2020	
Olaparib mean score	-2.00
Olaparib Bayes Factor (L,R)	676.34
Niraparib mean score	-2.74
Niraparib Bayes Factor (L,R)	2094.50
Rucaparib mean score	-2.7
Rucaparib Bayes Factor (L,R)	700.99
CBDCA (carboplatin) mean score	-2.61
CBDCA (carboplatin) Bayes Factor (L,R)	187.98
Morahan et al., 2019	
Complementation	Poor
HDR capacity (%)	15
Cisplatin sensitivity (%)	36
Couch lab BRCA2 assays	
Hart et al., 2019	
HDR fold change	1.27
HDR classification	deleterious
Source study	Guidugli et al., 2018
Results from the assay run that quantified the effect of the protein change were also directly used in an earlier publication featuring the same assay (Guidugli et al., 2018). They are not independent assay observations.	

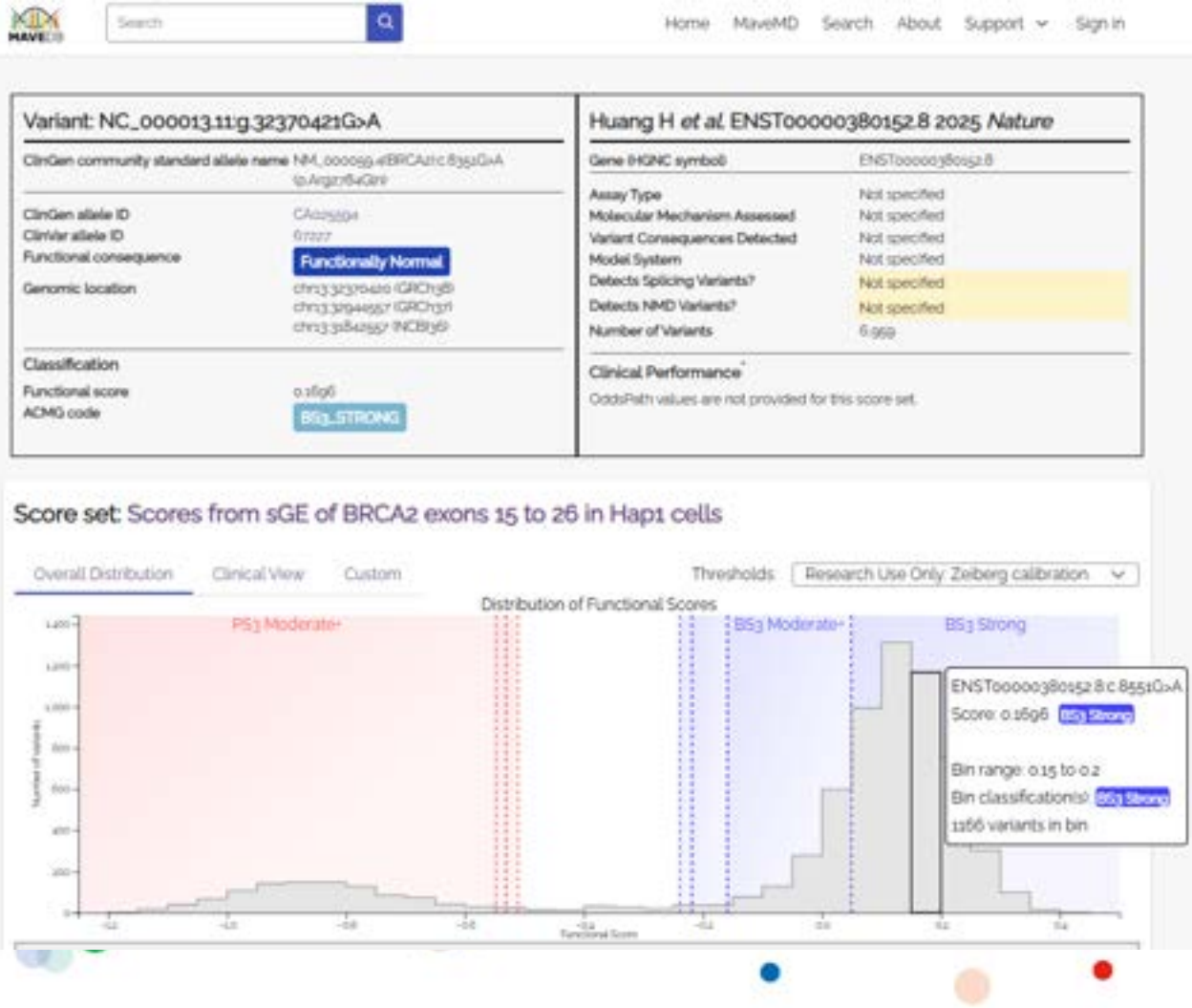
***BRCA2* c.8351G>A p.(Arg2784Gln): ENIGMA project**

- Found in trans with a *BRCA2* truncating variant in a Fanconi anaemia case
- Consistent with reduced penetrance
- ENIGMA project (being written up) confirms that penetrance for BC is moderate risk and OC risk is twice general population
- **WE NEED VALIDATED MAVE ASSAYS FOR KNOWN REDUCED PENETRANCE VARIANTS TO DEFINE THRESHOLDS VS CLINICAL DISEASE**



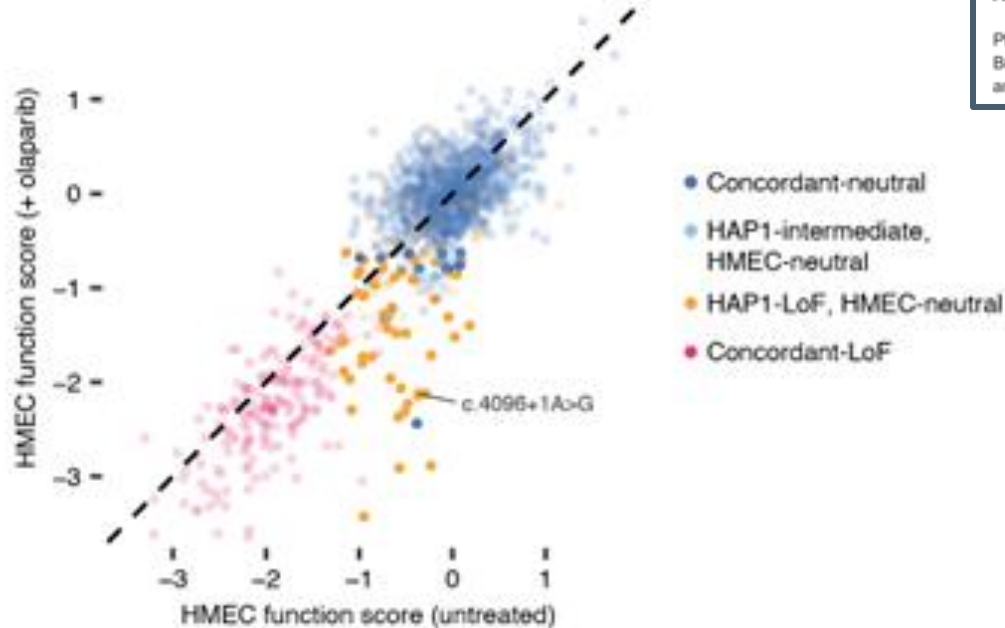
BRCA2 c.8351G>A
p.(Arg2784Gln)

New MAVE data
is benign!



Dace et al: Cell line affects *BRCA1* SGE scores

A



Saturation genome editing of *BRCA1* across cell types accurately resolves cancer risk

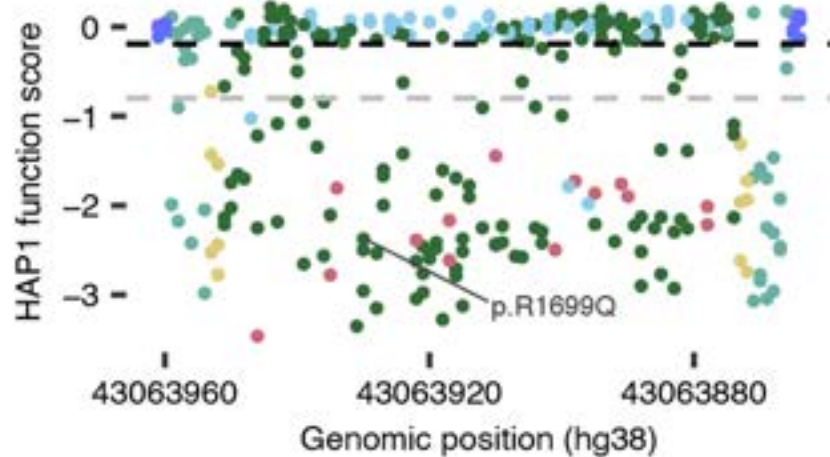
AUTHORS AND AFFILIATIONS

Phoebe Dace¹, Nicole M. Forrester¹, Maria Zanti², Laura Cubitt¹, Chloé Terwagne¹, Megan Buckley¹, Elke M. van Veen³, Thomas Stuart Wilson⁴, Paola Scaffidi^{4,5}, Kyriaki Michalidou² and Gregory M. Findlay¹

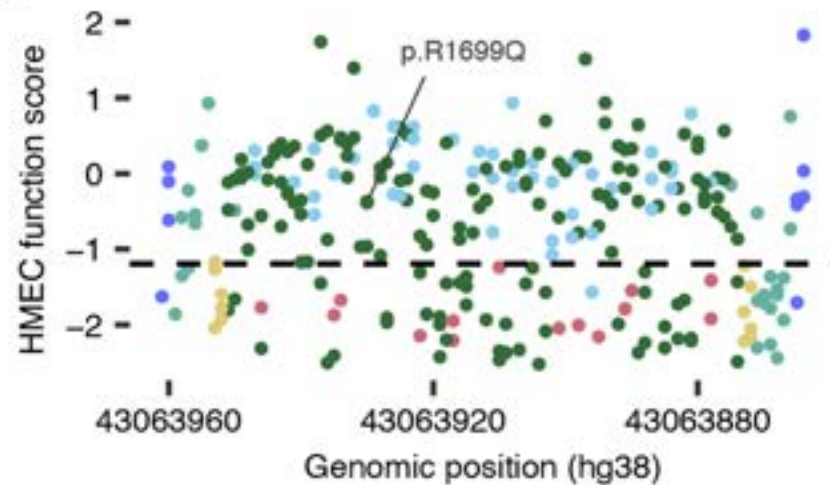
- 79.3% variants concordant
- HAP1 more sensitive to LOF than HMEC
- If LOF in both 12.97-fold risk increase
- If LOF in HAP1 and neutral in HMEC 3.28-fold risk increase
- Predicts reduced penetrance?
- Need to decide how we use this data clinically e.g. use combined datasets to accurately classify variants?

Our poster child: p.R1699Q

B



C



● Synonymous

● Nonsense

● Missense

● Splice region

● Intronic

● Canonical splice

Rare genetic variants in genes and loci linked to dominant monogenic developmental disorders cause milder related phenotypes in the general population

Authors

Rebecca Kingdom, Marcus Tuke, Andrew Wood,
Robin N. Beaumont, Timothy M. Frayling,
Michael N. Weedon, Caroline F. Wright

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Beaumont et al. *Genome Medicine* (2024) 16:64
<https://doi.org/10.1186/s13073-024-01333-4>

Genome Medicine

RESEARCH

Open Access

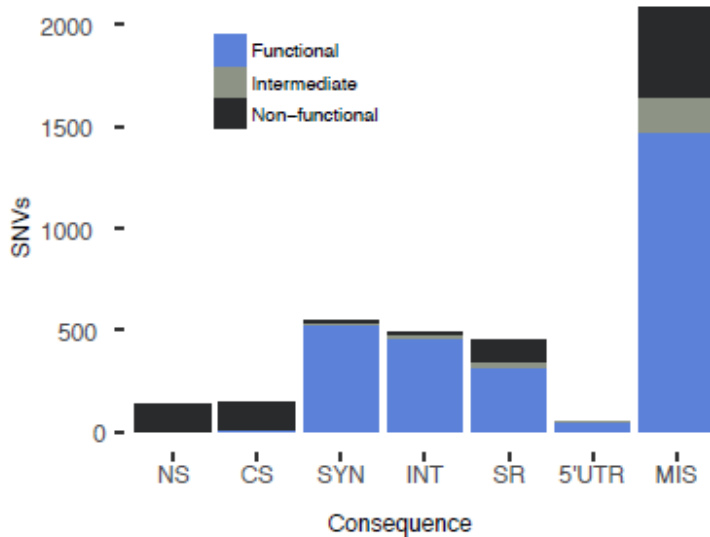
Clustering of predicted loss-of-function variants in genes linked with monogenic disease can explain incomplete penetrance

Robin N. Beaumont^{1*}, Gareth Hawkes¹, Adam C. Gunning^{1,2} and Caroline F. Wright^{1*}



Intermediate vs Indeterminate?

h



Reduced penetrance or indeterminate due to assay?

Figure 2 | Saturation genome editing enables functional classification of 3,893 *BRCAl* SNVs. **a**, HDR editing rates were calculated for each exon as the fraction of day 5 reads containing the SNV library's fixed synonymous variant (*i.e.* an 'HDR marker' edit). The average of two WT HAP1 replicates and two HAP1-Lig4KO replicates is plotted for comparison. (Asterisk denotes missing exon 22 data) **b**, The fraction of all possible SNVs scored is shown for each exon. SNVs were excluded mainly due to proximity to the HDR marker and/or poor sampling (Extended Data Fig. 6 and Methods). **c-f**, Reproducibility was assessed across all exon replicates (Extended Data Fig. 5). Measurements for exon 17 SNVs assayed in HAP1-Lig4KO cells are plotted to show correlations of day 5 frequencies (**c**, $p = 0.97$), day 11 over library ratios (**d**, $p = 0.95$), function scores (**e**, $p = 0.88$), and RNA expression scores (**f**, $p = 0.61$). **g**, A histogram of 3,893 SNV function scores (averaged across replicates and normalized across exons) shows how each category of mutation compares to the overall distribution. **h**, The number of SNVs within each category of mutation is plotted and colored by functional classification determined by SGE. (NS = nonsense, CS = canonical splice, SYN = synonymous, INT = intronic, SR = splice region, 5'UTR = 5' untranslated region, MIS = missense.)

Learnings from penetrance

- Penetrance of different genes is variable
- Penetrance of different phenotypes caused by same gene is variable
- Most penetrance estimates are based on families with strong family history (potential co-inheritance of modifiers) sometimes that led to discovery of that gene; they are often an over-estimate unless recent meta-analysis has been performed
- Penetrance of missense variants can vary from 0-100%; bio-physical size of change and location are critical
- Can be difficult to get enough patients and family members with same variant to definitively prove reduced penetrance – need expert VCEP (e.g. ENIGMA) to collect and collate these but not all gene have a VCEP





Mechanism of disease



Prevalence of loss-of-function, gain-of-function and dominant-negative mechanisms across genetic disease phenotypes

Received: 13 March 2025

Mihaly Badonyi[✉] & Joseph A. Marsh[✉]

Accepted: 8 August 2025



Loss-of-function AND dominant-negative effects

Table 3 | Top AR-AD phenotype pairs with high semantic similarity, where the dominant phenotype is likely to involve a dominant-negative effect

<i>Gene</i>	<i>Recessive phenotype</i>	<i>Dominant phenotype</i>	<i>Similarity</i>
POLG	Progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal recessive	Progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal dominant 1	0.775
AFG3L2	Spastic ataxia 5, autosomal recessive	Spinocerebellar ataxia 28	0.775
ACTA1	Congenital myopathy 2B, severe infantile, autosomal recessive	Congenital myopathy 2 C, severe infantile, autosomal dominant	0.763
ALDH18A1	Spastic paraplegia 9B, autosomal recessive	Spastic paraplegia 9 A, autosomal dominant	0.740
TTN	Muscular dystrophy, limb-girdle, autosomal recessive 10	Myopathy, myofibrillar, 9, with early respiratory failure	0.708
POLR3B	Leukodystrophy, hypomyelinating, 8, with or without oligodontia and/or hypogonadotropic hypogonadism	Charcot-Marie-Tooth disease, demyelinating, type 1I	0.700
HTRA1	Autosomal recessive cerebral arteriopathy with subcortical infarcts and leukoencephalopathy (CARASIL)	Cerebral arteriopathy, autosomal dominant, with subcortical infarcts and leukoencephalopathy, type 2	0.669
DEAF1	Dyskinesia, seizures, and intellectual developmental disorder	Vulto-van Silfout-de Vries syndrome	0.668
TWINK	Mitochondrial DNA depletion syndrome 7 (hepatocerebral type)	Progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal dominant 3	0.638
CLCN7	Osteopetrosis, autosomal recessive 4	Osteopetrosis, autosomal dominant 2	0.569
GHR	Laron syndrome	Growth hormone insensitivity, partial	0.521
LMNA	Mandibuloacral dysplasia	Restrictive dermopathy 2	0.511
SAMD9L	Myelodysplasia and leukaemia syndrome with monosomy 7	Ataxia-pancytopenia syndrome	0.510

Source data are provided as a Source Data file.

LOF and other non-LOF including GOF mechanism

Table 4 | The most different phenotype pairs within multi-phenotype disease genes by the mLOF score

Gene	LOF-like phenotype	mLOF	non-LOF-like phenotype	mLOF
SMCHD1	Facioscapulohumeral muscular dystrophy 2	0.606	Bosnia-Herzegovina Microphthalmia Syndrome	0.214
KRAS	Carpenter-Roeschke syndrome 2	0.623	Multiple myeloma	0.296
TP53	Rapp-Hodgkin syndrome	0.653	Split-Hand/Foot malformation 4	0.329
PPTB	Retinitis pigmentosa	0.656	Retinitis pigmentosa 13	0.339
SMARCB1	Schwannomatosis	0.65	Coffin-Siris syndrome 3	0.318
GRAS	Pseudopseudohypoparathyroidism	0.659	Pituitary adenoma 3, multiple types, somatic	0.346
BRAF	Neonan syndrome 1	0.632	Multiple myeloma	0.321
ABCG1	Cholestasis, progressive familial intrahepatic 2	0.638	Cholestasis, intrahepatic, of pregnancy 3	0.33
CAV3	Long QT syndrome 9	0.598	Rippling muscle disease	0.294
MTOR	Renal cell carcinoma, papillary, 1, familial and somatic	0.619	CIBAI3D syndrome	0.32
ADCY3	Dyskinesia with orofacial involvement, autosomal recessive	0.645	Dyskinesia, familial, with facial myokymia	0.364
ABR1	Epileptic encephalopathy, early infantile, 29	0.644	Charcot-Marie-Tooth disease, axonal, type 2N	0.365
LRP6	Tooth agenesis, selective, 7	0.656	Coronary artery disease, autosomal dominant 2	0.319
APMT	Deafness, X-linked 8	0.585	Spondylocraniofacial dysplasia, X-linked, with mental deterioration	0.375
ABR	Mental retardation, X-linked, with or without seizures, ABR-related	0.637	Corpus callosum, agenesis of, with abnormal genitalia	0.366
LMNGR	Hypergonadotropic hypogonadism	0.567	Precocious puberty, male	0.301
TRAPN1	Pernault syndrome 3	0.61	Progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal dominant 3	0.348
ACTA1	Congenital myopathy 28, severe infantile, autosomal recessive	0.656	Congenital myopathy 2 C, severe infantile, autosomal dominant	0.395
TP53	Breast cancer	0.648	Medulloblastoma	0.394
MPG2	Marfan dystrophy, spheriform, 5	0.648	Marfan dystrophy, spheriform, 2	0.395
HBA2	Hemoglobinemia	0.651	Methemoglobinemia, Alpha type	0.406
SLC32A1	Generalized epilepsy with febrile seizures plus, type 12	0.641	Developmental and epileptic encephalopathy 118	0.397
PCNA4	Congenital myopathy 22 A, classic	0.62	Paramyotonia congenita	0.376
EXT2	Exostoses, multiple, type II	0.633	Ovarian cancer	0.399
REF3L4	BAF300/INO syndrome	0.656	Ovarian cancer	0.422
POLR3B	Leukodystrophy, hypomyelinating, 8, with or without oligodendrocyte and/or hypogonadotropic hypogonadism	0.607	Charcot-Marie-Tooth disease, demyelinating, type 1	0.385
APCE	Lipoprotein glomerulopathy	0.539	Hyperproliferative anemia, type II	0.336
APRG	Acromesomelic dysplasia, Maroteaux type	0.656	Short stature with nonspecific skeletal abnormalities	0.395
PRNP	Gerstmann-Sträussler disease	0.552	Creutzfeldt-Jakob disease	0.331
FLNA	Cardiac valvular dysplasia, X-linked	0.652	Otopalatodigital syndrome, type I	0.432
TTN	Cardiomyopathy, dilated, 1 G	0.564	Myopathy, myofibrillar, 9, with early respiratory failure	0.395
SDHD	Mitochondrial complex II deficiency, nuclear type 3	0.634	Pheochromocytoma	0.407
BC1T1	Bertraphopathy, autosomal recessive	0.598	Vincoretrochoroidopathy	0.362
MECP2	Ret syndrome	0.655	Mental retardation, X-linked, syndromic 13	0.399
TPH4	Hypothyroidism, congenital, nongalactose, 1	0.594	Ovarian cancer	0.379
TRPV4	Metastatic dysplasia	0.568	Hereditary motor and sensory neuropathy, type IC	0.393

Source data are provided as a Source Data file.

CARD11 7:2906141-3044228

Reverse strand gene: caspase recruitment domain family member 11

Also known as: **CARMA1, BIMP3, ENSG00000198286**

Function: "Adapter protein that plays a key role in adaptive immune response by transducing the activation of NF-kappa-B downstream of T-cell receptor (TCR) and B-cell receptor (BCR) engagement (PubMed:11278692, PubMed:11356195, PubMed:12356734). Transduces signals downstream TCR or BCR ..." [Show more](#) Source: [UniProt](#)

DECIPHER holds no open-access sequence variants in this gene

[Overview](#) [Matching patient variants 144](#) [Matching DDD research variants 0](#) [Phenotypes](#) [Phenotype browser](#) [Expression](#) [Transcripts 12](#) [Browser](#)

[Clinical](#) [Protein / Genomic](#)

Gene/disease association

Gene2Phenotype ?

-

OMIM ?

607210

Morbid ?

- B-cell expansion with NFKB and T-cell anergy (Autosomal dominant)
- Immunodeficiency 11A (Autosomal recessive)
- Immunodeficiency 11B with atopic dermatitis (Autosomal dominant)

GeneReviews ?

-

Gene predictive scores

Probability of loss-of-function intolerance (pLI)

1.00

?

Loss-of-function observed/expected (LOEUF)

0.36

?

Heterozygous loss-of-function intolerance (sHet)

0.044

?

Probability of haploinsufficiency (pHaplo)

0.98

?

Probability of triplosensitivity (pTriplo)

0.16

?

Missense intolerance (Missense Z score)

3.73

?

Protein predictive scores

Probability of dominant-negative mechanism (pDN)

0.425

? Literature support

Probability of gain-of-function mechanism (pGOF)

0.475

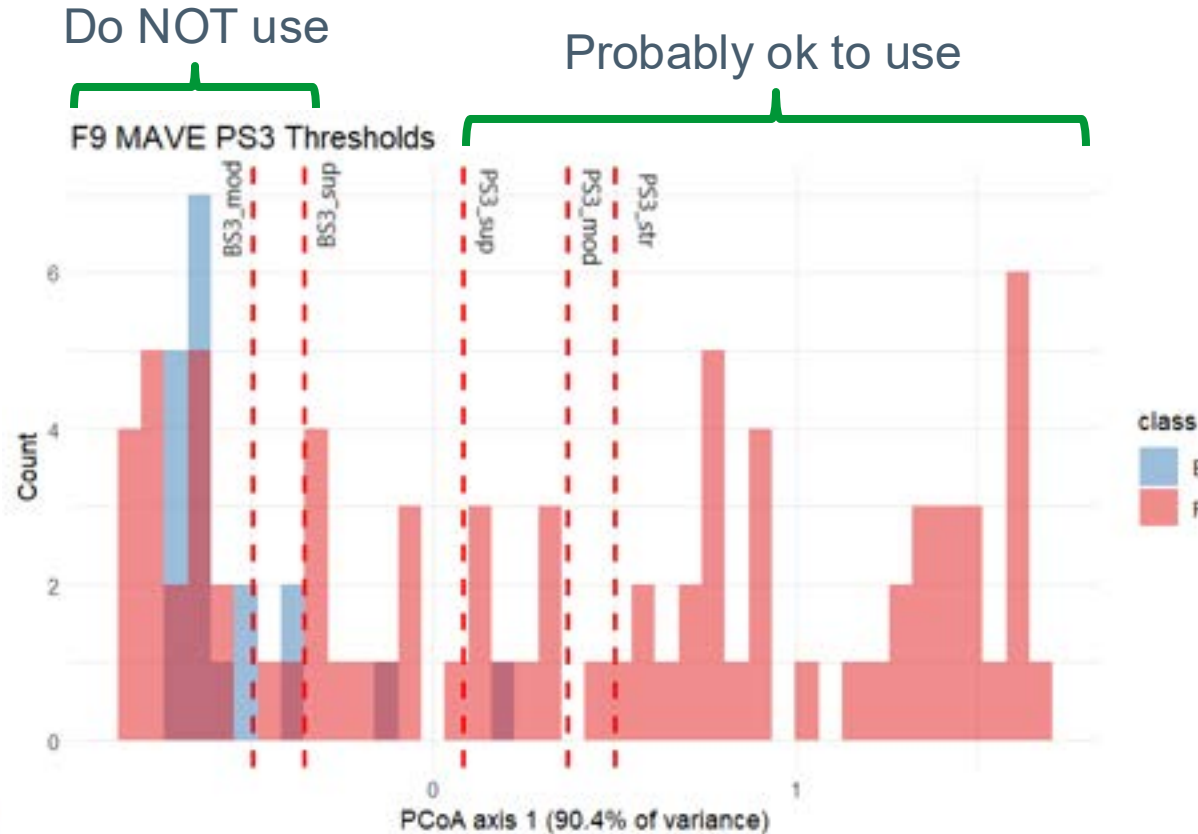
? Literature support

Probability of loss-of-function mechanism (pLOF)

0.634

?

Whole gene MAVE and different protein functional domains



Learnings from mechanism of disease

- Many genes have multiple mechanisms of disease
 - Can be different depending on mode of inheritance e.g. *COL1A2* LOF variants cause autosomal recessive Ehlers-Danlos Syndrome and Dominant Negative variants cause autosomal dominant Osteogenesis Imperfecta
 - Can be different depending on location in gene e.g. LOF at end of gene can cause gain-of-function disease (*NOTCH2* and *NOTCH3*)
 - *CARD11* gene has 3 different mechanisms of disease associated with 3 different phenotypes
- Many genes have different functions and depending on assay you may be able to use evidence towards pathogenicity but not benignity if the assay does not cover that mechanism e.g. Jia et al assay for splicing; F9 assay example using acmgscaler





